MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, JK, DIGITAL, KEY CODE FRE

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic JK flip-flop module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1239 failures per million operating hours for class I or 0.1700 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance and requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be FRE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/315-1 for class I and M28787/315-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the oerating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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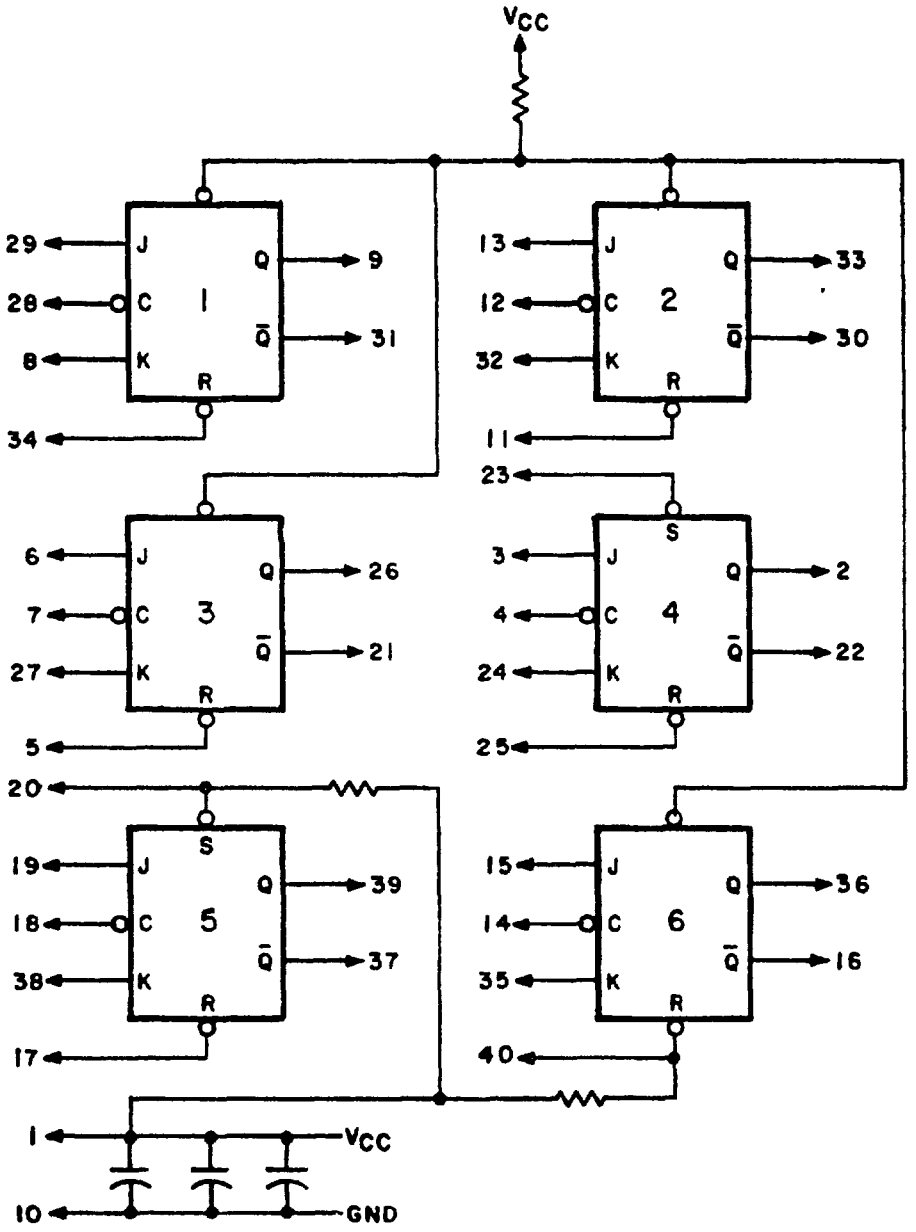


FIGURE 1. FRE module function diagram, JK flip-flop.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/ -3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B	V_{CC} stress	IV	S	← The module shall pass the remaining tests →										
C1-C26	Breakdown current	V	S	0	1	0	1	0	1	0	1	0	1	mA
D1-D12	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E26	Clamp diode voltage	VII	X	0.0	1.2	0.0	1.2	0.0	1.2	0.0	1.2	0.0	1.2	V
F1-F14	Functional	VIII	X	← See table VIII →										
G1-G12	Output voltage, low level	IX	X	0	500	0	500	0	500	0	500	0	500	mV
H1-H12	Output voltage, high level	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1,J3, J5,J7, J9,J11	Input current, high level	X	X	0	50	0	50	0	50	0	50	0	50	µA
J2,J4, J6,J8, J10, J12-J24	Input current, high level	X	X	0	100	0	100	0	100	0	100	0	100	µA
J25	Input current, high level	X	X	-166	-295	-166	-295	166	-295	-166	-295	-166	-295	µA
J26	Input current, high level	X	X	-467	-626	-467	-626	-467	-626	-467	-626	-467	-626	µA
K1-K12	Input current, low level	XI	X	0.00	-1.77	0.00	-1.77	0.00	-1.77	0.00	-1.77	0.00	-1.77	mA
K13-K23. K25	Input current, low level	XI	X	0.00	-7.73	0.00	-7.73	0.00	-7.73	0.00	-7.73	0.00	-7.73	mA

See footnotes at end of table.

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TABLE I. Module electrical requirements - continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K24	Input current, low level	XI	X	0.00	-8.23	0.00	-8.23	0.00	-8.23	0.00	-8.23	0.00	-8.23	mA
K26	Input current, low level	XI	X	0.00	-8.93	0.00	-8.93	0.00	-8.93	0.00	-8.93	0.00	-8.93	mA
L1,L2	Power supply current	XII	X	5	150	5	150	5	150	5	150	5	150	mA
M1-M12	Propagation delay time (t_{PLH}), C_L = 50 pF	XIII	S	2	11	2	13	2	13	2	17	2	15	ns
M13-M24	Propagation delay time (t_{PLH}), C_L = 150 pF	XIII	Q	2	21	2	23	2	23	2	27	2	25	ns
N1-N12	Propagation delay time (t_{PHL}), C_L = 50 pF	XIII	S	2	11	2	13	2	13	2	17	2	15	ns
N13-N24	Propagation delay time (t_{PHL}), C_L = 150 pF	XIII	Q	2	21	2	23	2	23	2	27	2	25	ns
P1-P24	Setup time	XIV	S	900	1,100	900	1,100	900	1,100	900	1,100	900	1,100	ns
Q1-Q24	Hold time	XV	S	900	1,100	900	1,100	900	1,100	900	1,100	900	1,100	ns
R1,R2, R5-R10	Propagation delay time (t_{PLH}), C_L = 50 pF	XVI	S	2	11	2	13	2	13	2	17	2	15	ns
R3,R4, R11-R16	Propagation delay time (t_{PLH}), C_L = 150 pF	XVI	Q	2	21	2	23	2	23	2	27	2	25	ns
S1,S2, S5-S10	Propagation delay time (t_{PHL}), C_L = 50 pF	XVI	S	2	11	2	13	2	13	2	17	2	15	ns
S3,S4, S11-S16	Propagation delay time (t_{PHL}), C_L = 150 pF	XVI	Q	2	21	2	23	2	23	2	27	2	25	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		85° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T1-T12	Maximum clock frequency check	XVII	S	25	33	25	33	25	33	25	33	25	33	ns
U1-U12	Output crosstalk, high level	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV
V1-V12	Output crosstalk, low level	XVIII	Q	0	500	0	500	0	500	0	500	0	500	mV
W	Power supply transient amplitude	XIX	Q	0	400	0	400	0	400	0	400	0	400	mV
X	Power supply filter capacitance	XX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
Y1-Y12	Input capacitance	XXI	Q	2	18	2	18	2	18	2	18	2	18	pF
Y13-Y26	Input capacitance	XXI	Q	2	25	2	25	2	25	2	25	2	25	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 85° C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	825	825	mW
Input current			
Low level (V_{IL} = 0 V			
J, K	-1.77 max	-1.77 max	mA
C	-4.41 max	-4.41 max	mA
Low level (V_{IL} = 0.5 V)			
IR - 5R, 4S	-7.73 max	-7.73 max	mA
5S	-8.23 max	-8.23 max	mA
6R	-8.93 max	-8.93 max	mA
Input current			
High level (V_{IL} = 5.5 V)	1.0 max	1.0 max	mA
High level (V_{IH} = 2.7 V)			
J	50 max	50 max	µA
K, C, IR-5R, 4S	100 max	100 max	µA
5S	282 max	282 max	µA
6R	626 max	626 max	µA
Output drive current			
Low level (V_{OL} = 0.5 V)	20 max	20 max	mA
High level (V_{OH} = 2.5 V)	1 max	1 max	mA
Setup and hold time			
t_{Hold}	2 max	2 max	ns
t_{Setup}	9 max	9 max	ns

6.5 Input to output relationship. Each JK flip-flop module has the input to output relationship of table II for positive logic.

6.6 Parallel outputs. Paralleling of outputs is not permissible.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N379)

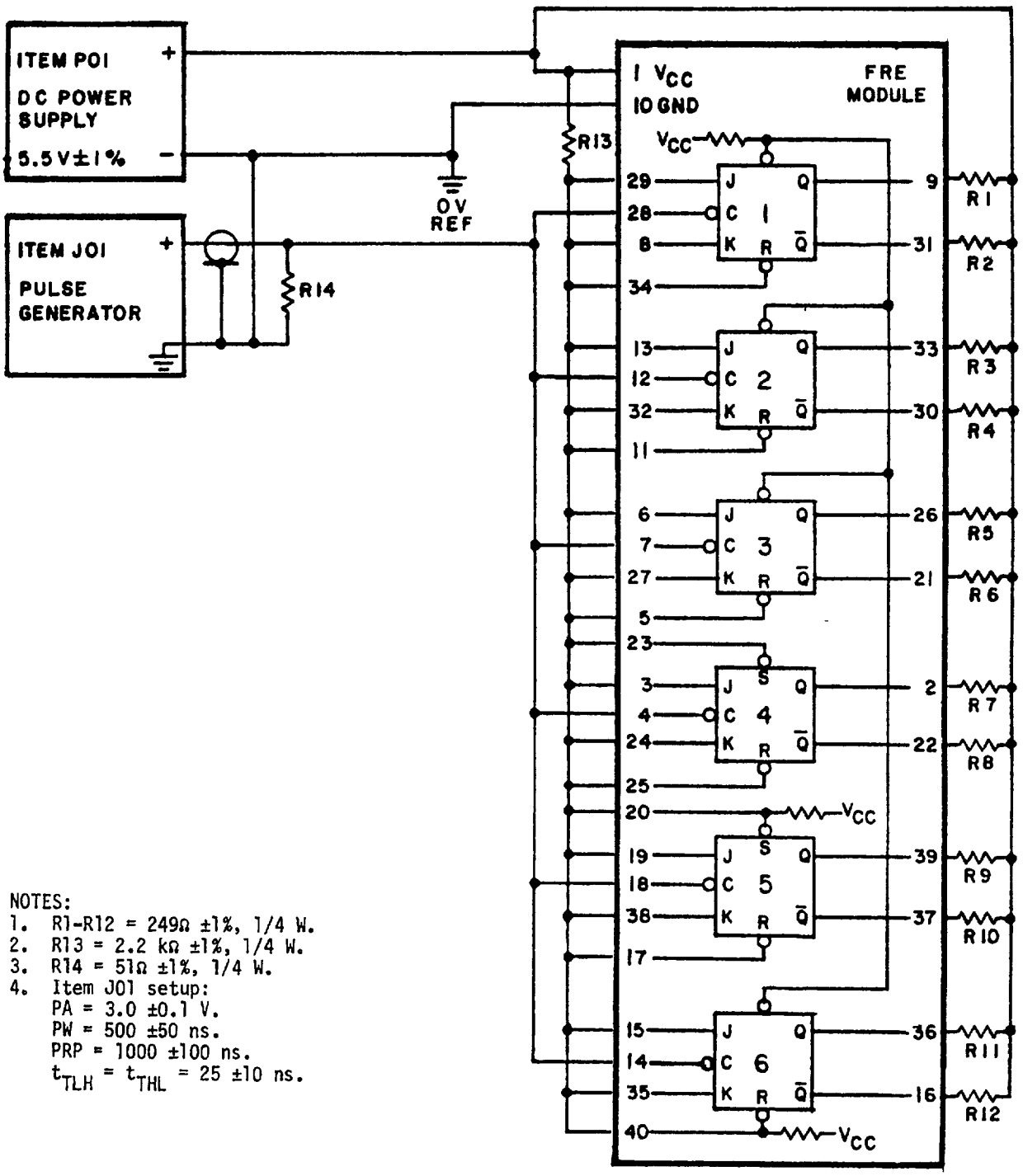
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TABLE II. Input to output relationship. 1/ - 6/

Inputs					Outputs	
S	R	C	J	K	Qn	$\overline{Q}$n
L	H	X	X	X	H	L
H	L	X	X	X	L	H
L	L	X	X	X	H'	H'
H	H	↓	L	L	Qo	$\overline{Q}$o
H	H	↓	H	L	H	L
H	H	↓	L	H	L	H
H	H	↓	H	H	Toggle	
H	H	H	X	X	Qo	$\overline{Q}$o

1/ H' = unstable state.
2/ X = indifferent level.
3/ H = high level.
4/ L = low level.
5/ Qn = output Q before falling edge of clock signal.
6/ Qo, $\overline{Q}$o = level of Q/$\overline{Q}$ before the indicated steady state input conditions were established.

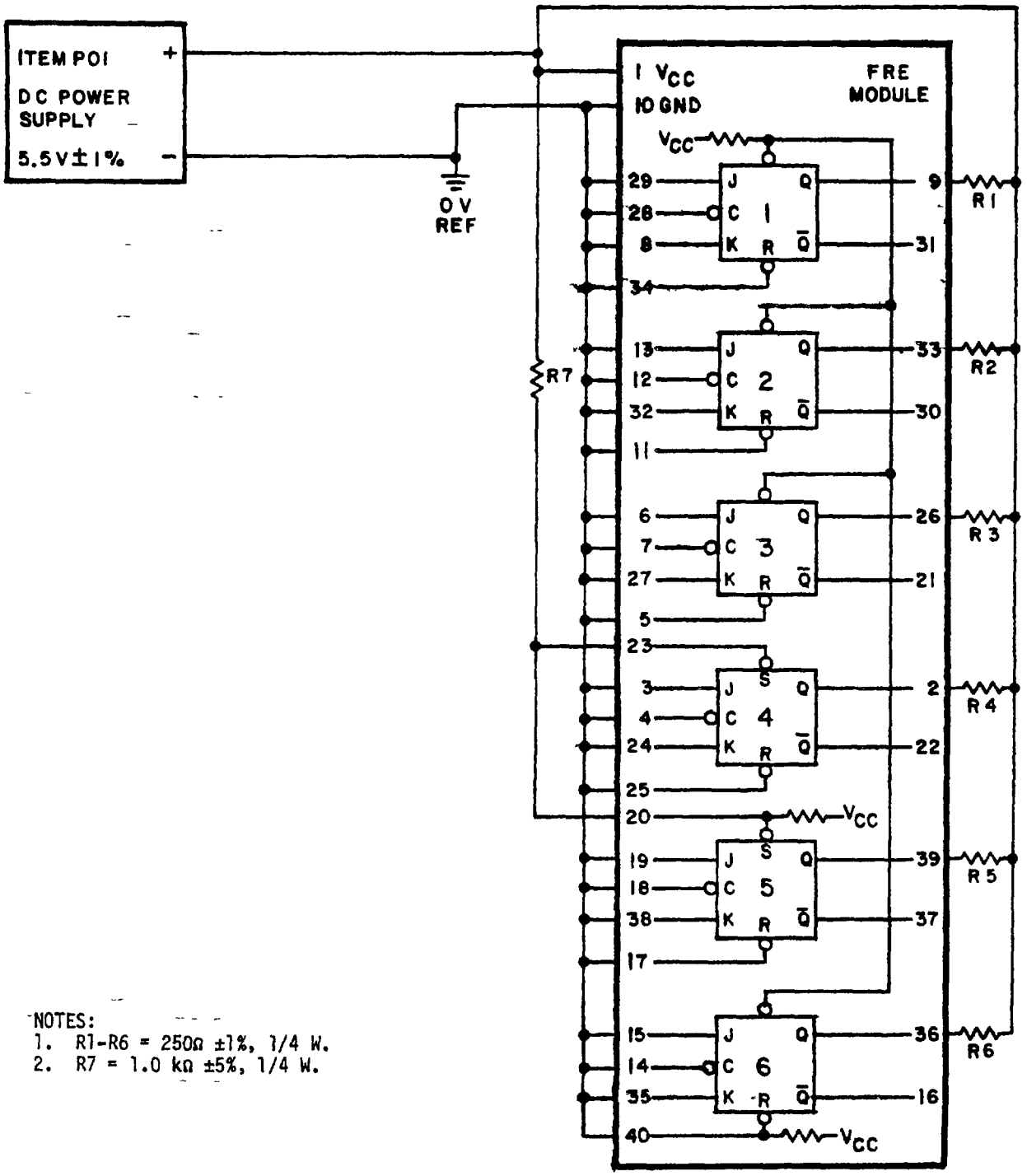
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NOTES:
1. R1-R12 = 249Ω ±1%, 1/4 W.
2. R13 = 2.2 kΩ ±1%, 1/4 W.
3. R14 = 51Ω ±1%, 1/4 W.
4. Item J01 setup:
   PA = 3.0 ±0.1 V.
   PW = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   $t_{TLH} = t_{THL} = 25 \pm 10$ ns.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1-R6 = 250Ω ±1%, 1/4 W.
2. R7 = 1.0 kΩ ±5%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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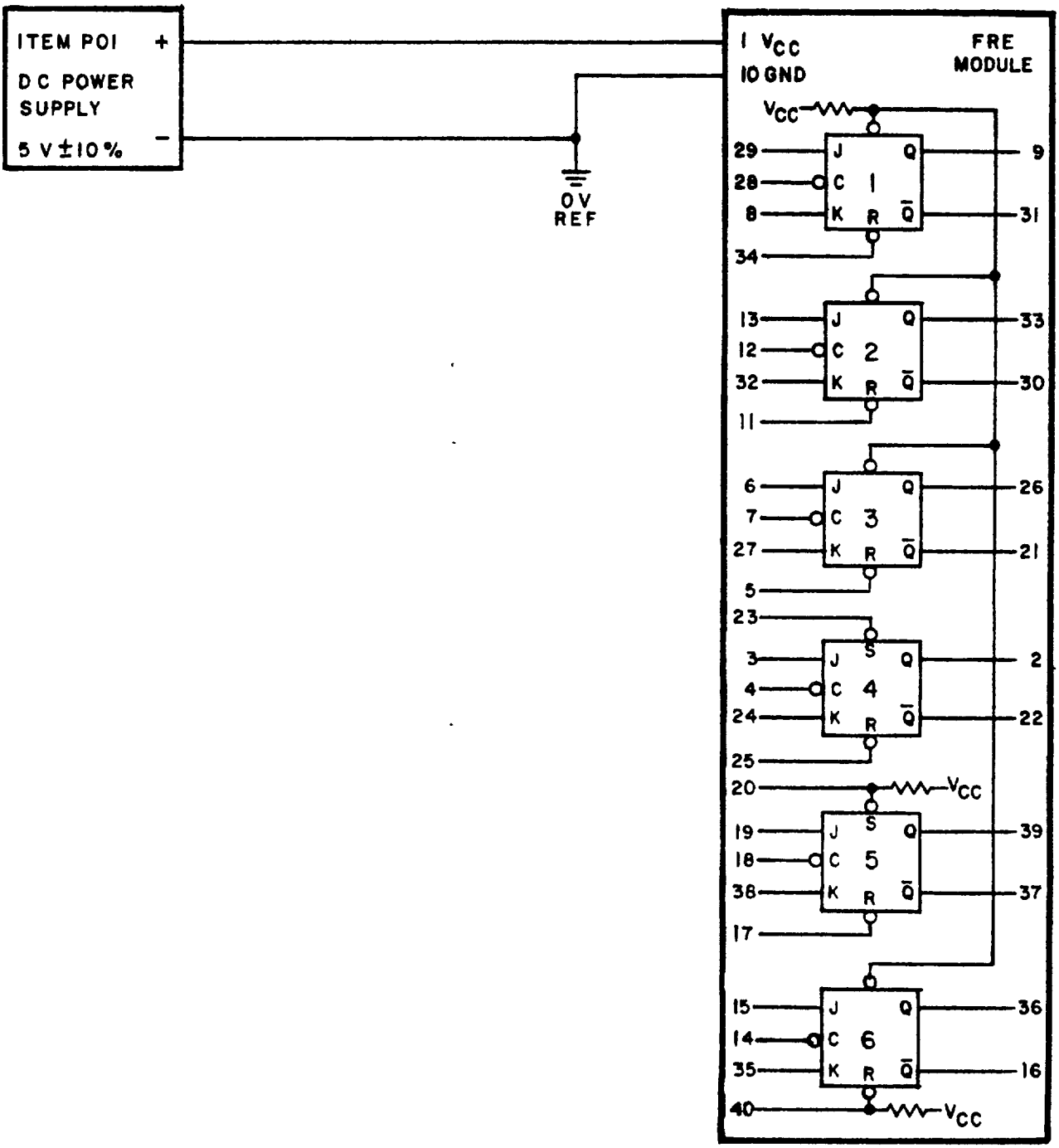


FIGURE 4. Vibration test circuit.

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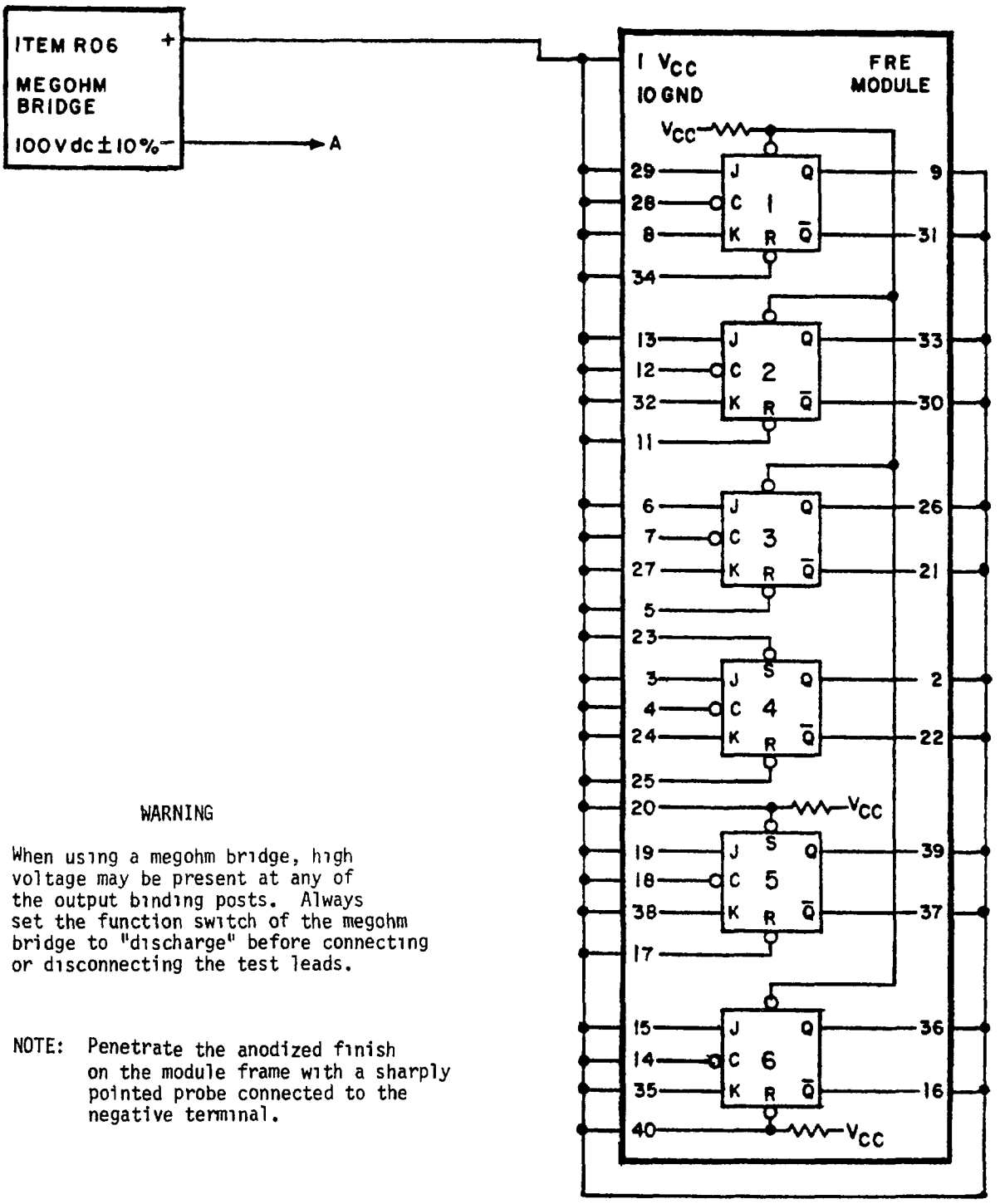
TABLE III. Isolation test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A	Isolation	Frame	10	∞	MΩ	5

TABLE IV. V_{CC} stress test.

Test condition	Parameter	Initial limits 25° ±5°C ambient		Unit	Test figure
		Min	Max		
B	V_{CC} stress	The module shall pass the remaining tests			6

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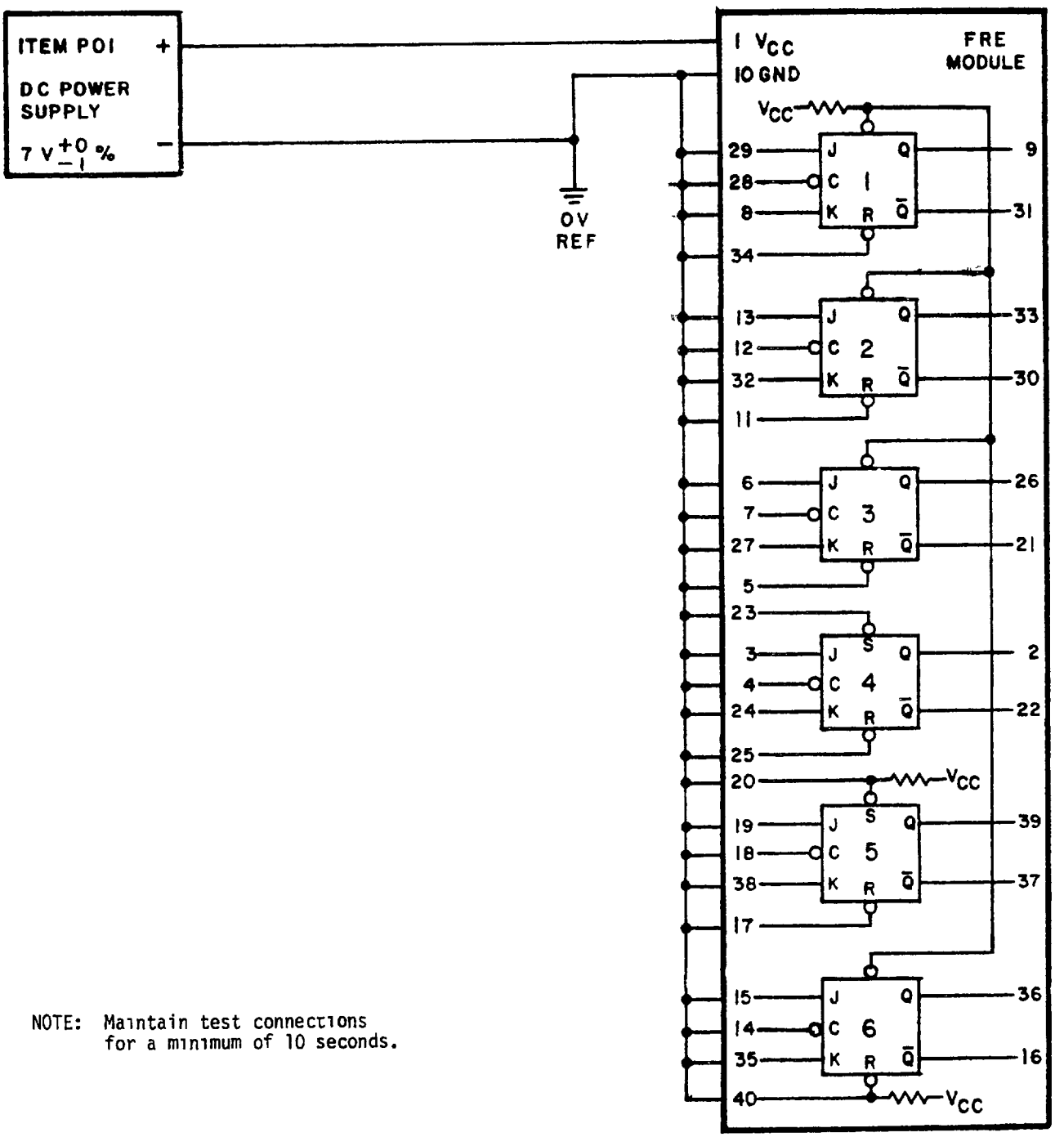
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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NOTE: Maintain test connections for a minimum of 10 seconds.

FIGURE 6. V_{CC} stress test circuit for table IV.

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TABLE V. Breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1J	Breakdown current	28,8,34,13,12,32,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17, 15,14,35,40	29	0	1	mA	7
C2	1C		29,8,34,13,12,32,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	28				
C3	1K		29,28,34,13,12,32,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	8				
C4	1R		29,28,8,13,12,32,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	34				
C5	2J		29,28,8,34,12,32,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	13				
C6	2C		29,28,8,34,13,32,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	12				
C7	2K		29,28,8,34,13,12,11, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	32				

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TABLE V. Breakdown current test - continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C8	2R	Breakdown current	29,28,8,34,13,12,32, 6,7,27,5,23,3,4,24, 25,20,19,18,38,17, 15,14,35,40	11	0	1	mA	7
C9	3J		29,28,8,34,13,12,32, 11,7,27,5,23,3,4,24, 25,20,19,18,38,17, 15,14,35,40	6				
C10	3C		29,28,8,34,13,12,32, 11,6,27,5,23,3,4,24, 25,20,19,18,38,17, 15,14,35,40	7				
C11	3K		29,28,8,34,13,12,32, 11,6,7,5,23,3,4,24, 25,20,19,18,38,17, 15,14,35,40	27				
C12	3R		29,28,8,34,13,12,32, 11,6,7,27,23,3,4,24, 25,20,19,18,38,17,15, 14,35,40	5				
C13	4S		29,28,8,34,13,12,32, 11,6,7,27,5,3,4,24, 25,20,19,18,38,17,15, 14,35,40	23				
C14	4J		29,28,8,34,13,12,32, 11,6,7,27,5,23,4,24, 25,20,19,18,38,17,15, 14,35,40	3				
C15	4C		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,24, 25,20,19,18,38,17,15, 14,35,40	4				

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TABLE V. Breakdown current test - continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C16	4K	Breakdown current	29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 25,20,19,18,38,17, 15,14,35,40	24	0	1	mA	7
C17	4R		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,20,19,18,38,17, 15,14,35,40	25				
C18	5S		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,19,18,38,17,15, 14,35,40	20				
C19	5J		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,18,38,17,15, 14,35,40	19				
C20	5C		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,38,17,15, 14,35,40	18				
C21	5K		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,18,17,15, 14,35,40	38				
C22	5R		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,18,38,15, 14,35,40	17				
C23	5J		29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,18,38,17, 14,35,40	15				

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TABLE V. Breakdown current test - continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C24	6C	Breakdown current	29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,18,38, 17,15,35,40	14	0	1	mA	7
C25	6K	Breakdown current	29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,18,38, 17,15,14,40	35	0	1	mA	7
C26	6R	Breakdown current	29,28,8,34,13,12,32, 11,6,7,27,5,23,3,4, 24,25,20,19,18,38,17, 15,14,35	40	0	1	mA	7

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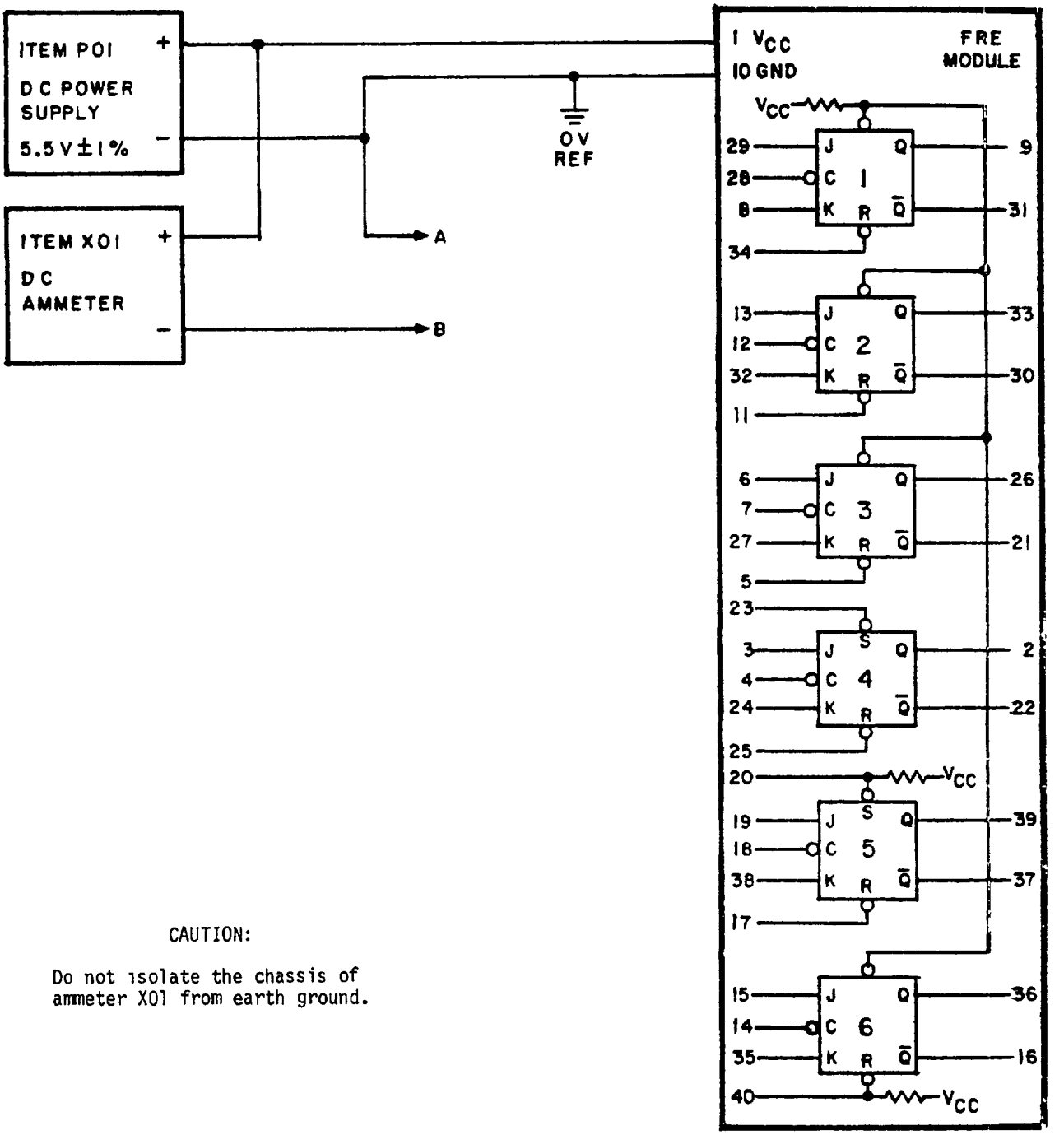


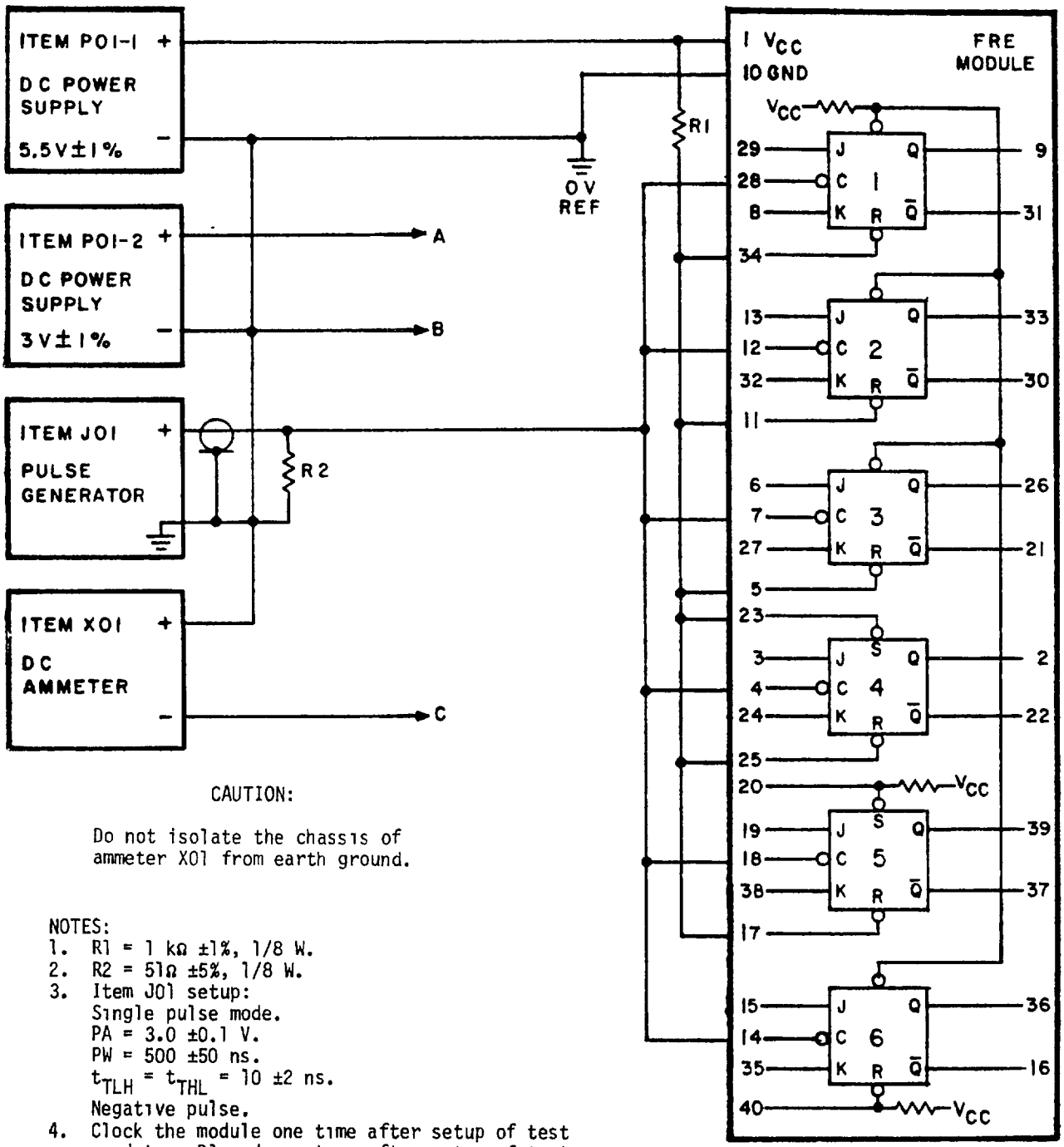
FIGURE 7. Breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	1Q Output short circuit current	29,13,6,3,19,15	8,32,27,24,38,35	9	-40	-100	mA	8
D2	2Q			33				
D3	3Q			26				
D4	4Q			2				
D5	5Q			39				
D6	6Q			36				
D7	1$\overline{Q}$	8,32,27,24,38,35	29,13,6,3,19,15	31				
D8	2$\overline{Q}$			30				
D9	3$\overline{Q}$			21				
D10	4$\overline{Q}$			22				
D11	5$\overline{Q}$			37				
D12	6$\overline{Q}$			16				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = 51Ω ±5%, 1/8 W.
3. Item J01 setup:
   Single pulse mode.
   PA = 3.0 ±0.1 V.
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
   Negative pulse.
4. Clock the module one time after setup of test condition D1 and one time after setup of test condition D7.

FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Clamp diode voltage test.

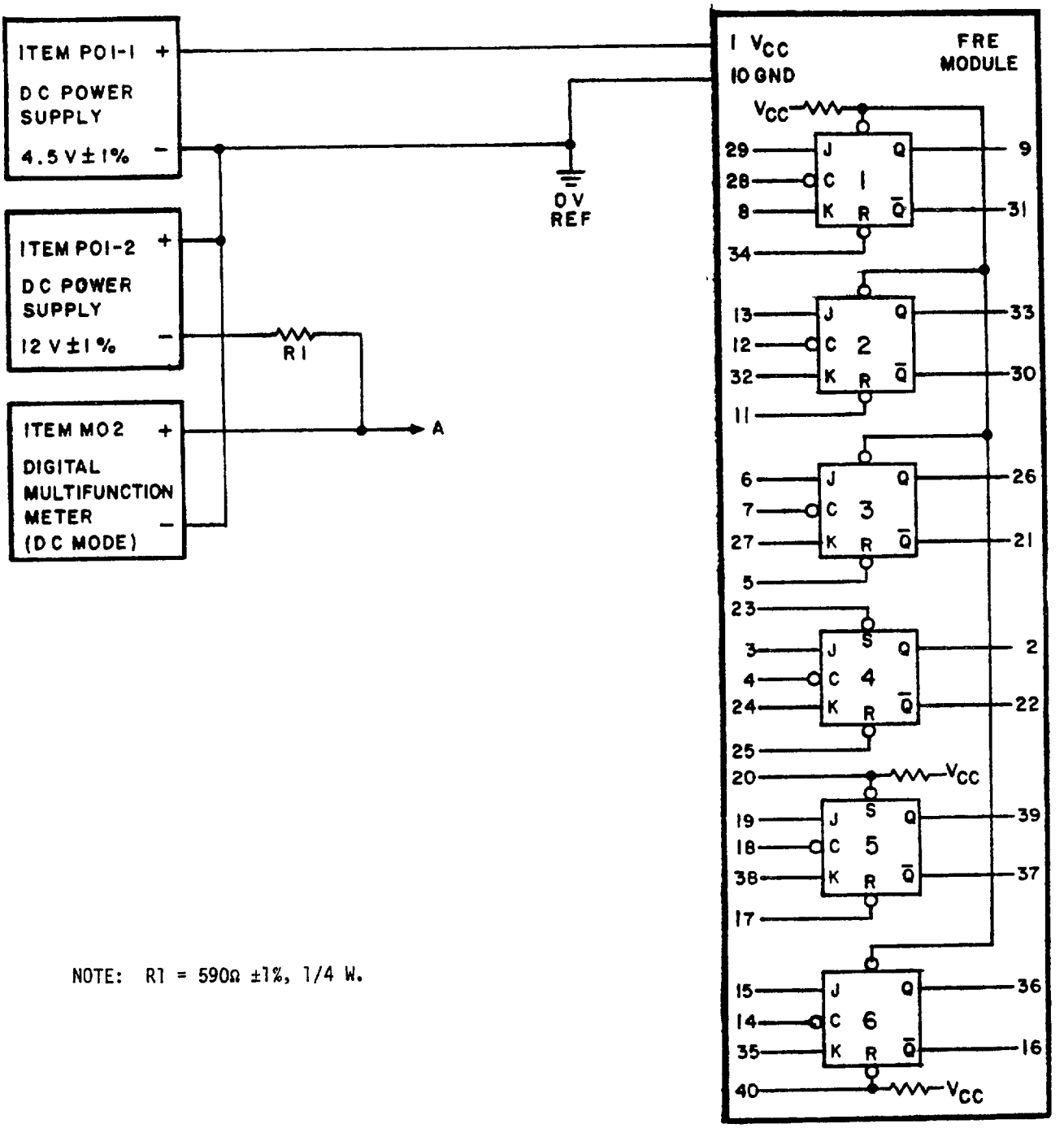
Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	1J Clamp diode voltage	29	0.0	1.2	V	9
E2	1C	28				
E3	1K	8				
E4	1R	34				
E5	2J	13				
E6	2C	12				
E7	2K	32				
E8	2R	11				
E9	3J	6				
E10	3C	7				
E11	3K	27				
E12	3R	5				
E13	4S	23				
E14	4J	3				
E15	4C	4				
E16	4K	24				
E17	4R	25				
E18	5S	20				
E19	5J	19				
E20	5C	18				

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TABLE VII. Clamp diode voltage test - continued.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E21	5K Clamp diode voltage	38	0.0	1.2	V	9
E22	5R	17	↓	↓	↓	↓
E23	6J	15				
E24	6C	14				
E25	6K	35				
E26	6R ↓	40				

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NOTE: R1 = 590Ω ±1%, 1/4 W.

FIGURE 9. Clamp diode voltage test circuit for table VII.

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TABLE VIII. Functional test.

Test condition 3/	Parameter	Connect				Outputs												Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	$1\overline{Q}$	$2\overline{Q}$	$3\overline{Q}$	$4\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	Min	Max		
F1	R Functional	23	29,8,34, 13,32,11, 6,27,5,3, 24,25,19, 38,17,15, 35,40	2/	28,12,7, 4,18,14	L	L	L	L	L	L	H	H	H	H	H	H	1/			10
	J = L																				
	K = L																				
	R = L																				
F2	R Functional	23,8,32, 27,24,38, 35	29,34,13, 11,6,5, 3,25,19, 17,15,40			L	L	L	L	L	L	H	H	H	H	H	H				
	J = L																				
	K = H																				
	R = L																				
F3	R Functional	23,29,8, 13,32,6, 27,3,24, 19,38,15, 35	34,11,5, 25,17,40			L	L	L	L	L	L	H	H	H	H	H	H				
	J = H																				
	K = H																				
	R = L																				
F4	R Functional	23,29,13, 6,3,19,15	8,34,32, 11,27,5, 24,25,38, 17,35,40			L	L	I	L	L	L	H	H	H	H	H	H				
	J = H																				
	K = L																				
	R = L																				
F5	Functional	29,8,34, 13,32,11, 6,27,5, 23,3,24, 25,19,38, 17,15,35	Open			H	H	H	H	H	H	L	L	L	L	L	L				
	J = H																				
	K = H flip-flop counter																				
	R = H																				

See footnotes at end of table.

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TABLE VIII. Functional test - continued.

Test condition 3/	Parameter	Connect				Outputs												Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	$1\overline{Q}$	$2\overline{Q}$	$3\overline{Q}$	$4\overline{Q}$	$5\overline{Q}$	$6\overline{Q}$	Min	Max		
F6	Functional J = H K = H flip-flop counter R = H	29,8,34,13,32,11,6,27,5,23,3,24,25,19,38,17,15,35	Open	2/	28,12,7,4,18,14	L	L	L	L	L	L	H	H	H	H	H	H	1/			10
F7	JK Functional J = H K = L R = H	29,34,13,11,6,5,23,3,25,19,17,15	8,32,27,24,38,35			H	H	H	H	H	H	L	L	L	L	L	L				
F8	Functional (without count) J = L K = L R = H	23,34,11,5,25,17,40,20	29,8,13,32,6,27,3,24,19,38,15,35			H	H	H	H	H	H	L	L	L	L	L	L				
F9	JK Functional J = L K = H R = H	23,8,34,32,11,27,5,24,25,38,17,35	29,13,6,3,19,15			L	L	L	L	L	L	H	H	H	H	H	H				
F10	Functional (without count) J = L K = L R = H	23,34,11,5,25,17	29,8,13,32,6,27,3,24,19,38,15,35			L	L	L	L	L	L	H	H	H	H	H	H				

See footnotes at end of table.

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TABLE VIII. Functional test - continued

Test condition 3/	Parameter	Connect				Outputs												Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	1Q	2Q	3Q	4Q	5Q	6Q	1$\overline{Q}$	2$\overline{Q}$	3$\overline{Q}$	4$\overline{Q}$	5$\overline{Q}$	6$\overline{Q}$	Min	Max		
F11	S Functional	25,17	23,3,24, 20,19,38	2/	4,18	L	L	L	H	H	L	H	H	H	L	L	H	1/			10
	J = L																				
	K = L																				
F12	S Functional	25,17,3, 24,19,38	23,20						H	H					L	L					
	J = H																				
	K = H																				
F13	S Functional	25,17, 3,19	23,20, 24,38						H	H					L	L					
	J = H																				
	K = L																				
F14	S Functional	25,17, 24,38	23,20, 3,19						H	H					L	L					
	J = L																				
	K = H																				

1/ Low level, 0.5 V, max; H = high level, 2.5 V, min.

2/ For test conditions F1 thru F14, check that the logic level complies with that specified in footnote 1, by verifying outputs using point C once the input conditions have been established.

3/ Test conditions F1 thru F14 shall be run sequentially, changing only the specified inputs

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TABLE IX. Output voltage (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	1Q	Output voltage, low level	8,34	29	9	28	0	500	mV	10
G2	2Q	↓	32,11	13	33	12	↓	↓	↓	↓
G3	3Q		27,5	6	26	7				
G4	4Q		24,25,23	3	2	4				
G5	5Q		38,17,20	19	39	18				
G6	6Q		35,40	15	36	14				
G7	$1\overline{Q}$		29,34	8	31	28				
G8	$2\overline{Q}$		13,11	32	30	12				
G9	$3\overline{Q}$		6,5	27	21	7				
G10	$4\overline{Q}$		23,3,25	24	22	4				
G11	$5\overline{Q}$		19,17	38	37	18				
G12	$6\overline{Q}$		15,40	35	16	14				

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TABLE IX. Output voltage (low-level and high-level) test - continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1Q	Output voltage, high level	29,34	8	9	28	2.5	4.5	V	10
H2	2Q	↓	13,11	32	33	12	↓	↓	↓	↓
H3	3Q		6,5	27	26	7				
H4	4Q		23,3,25	24	2	4				
H5	5Q		19,17,20	38	39	18				
H6	6Q		15,40	35	36	14				
H7	$1\overline{Q}$		8,34	29	31	28				
H8	$2\overline{Q}$		32,11	13	30	12				
H9	$3\overline{Q}$		27 5	6	21	7				
H10	$4\overline{Q}$		24,25,23	3	22	4				
H11	$5\overline{Q}$		38,17	19	37	18				
H12	$6\overline{Q}$		35,40	15	16	14				

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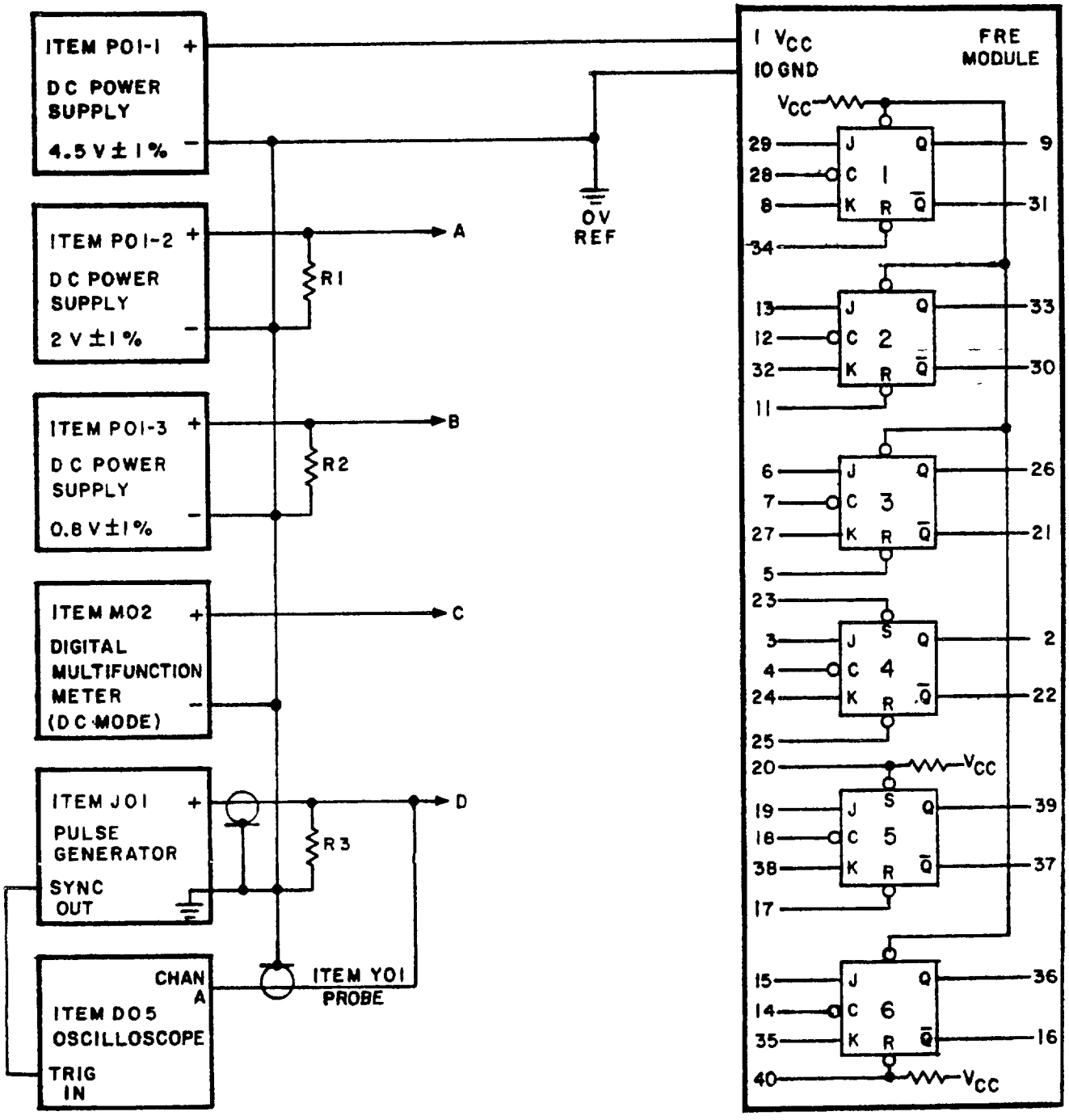
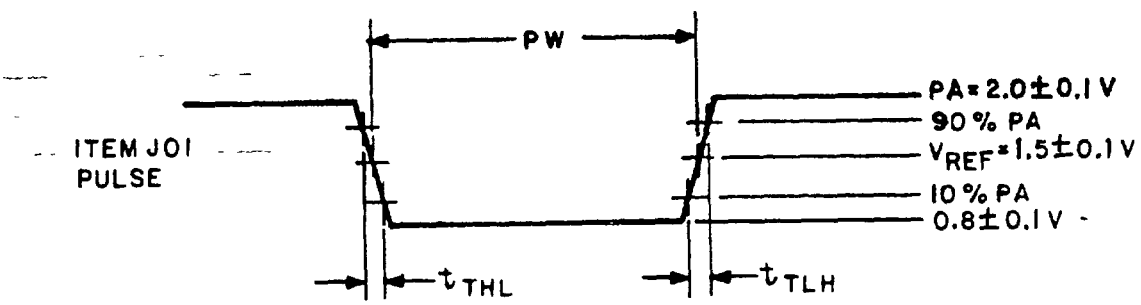


FIGURE 10. Functional and output voltage (low-level and high-level) test circuit for tables VIII and IX.

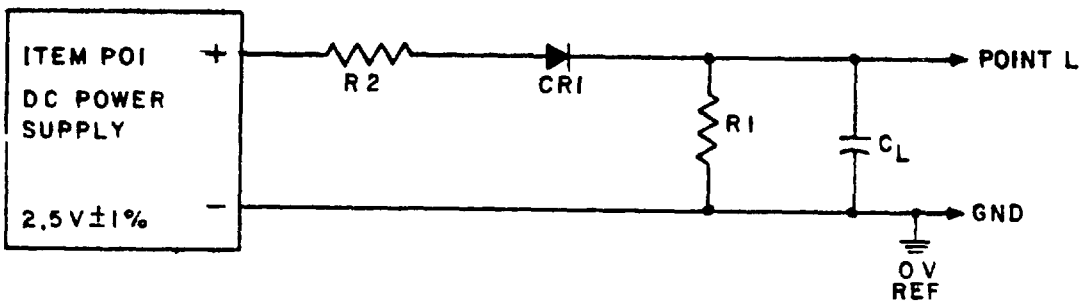
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 51.1Ω ±1%, 1/4 W.
3. Load Q and $\overline{Q}$ outputs with load as specified on figure 11 for test conditions G1-G12 and H1-H12.
4. Item J01 setup: (see above pulse)
   PW = 500 ±50 ns.
   $t_{TLH} = t_{THL}$ = 10 ±2 ns.
5. Apply one test pulse, then perform the measurement.

FIGURE 10. Functional and output voltage (low-level and high-level) test circuit for tables VIII and IX - Continued.

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NOTES:
1. R1 = 2.49 kΩ ±1%.
2. R2 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
3. CR1 = 1N3600 or equivalent.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions M1-M12, N1-N12, P1-P24, Q1-Q24, R1-R8, S1-S8, T1-T12, U1-U12 and V1-V12.
5. C_L = 150 ±15 pF (including probe and parasitic capacitance) for test conditions M13-M24, N13-N24, P13-P24, R9-R16 and S9-S16.
6. C_L is not required for test conditions G1-G12 and H1-H12.
7. Lead length from point L to test point shall be less than or equal to 2 inches (50.8 mm).

FIGURE 11. FRE logic load circuit.

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TABLE X. Input current (high-level) test.

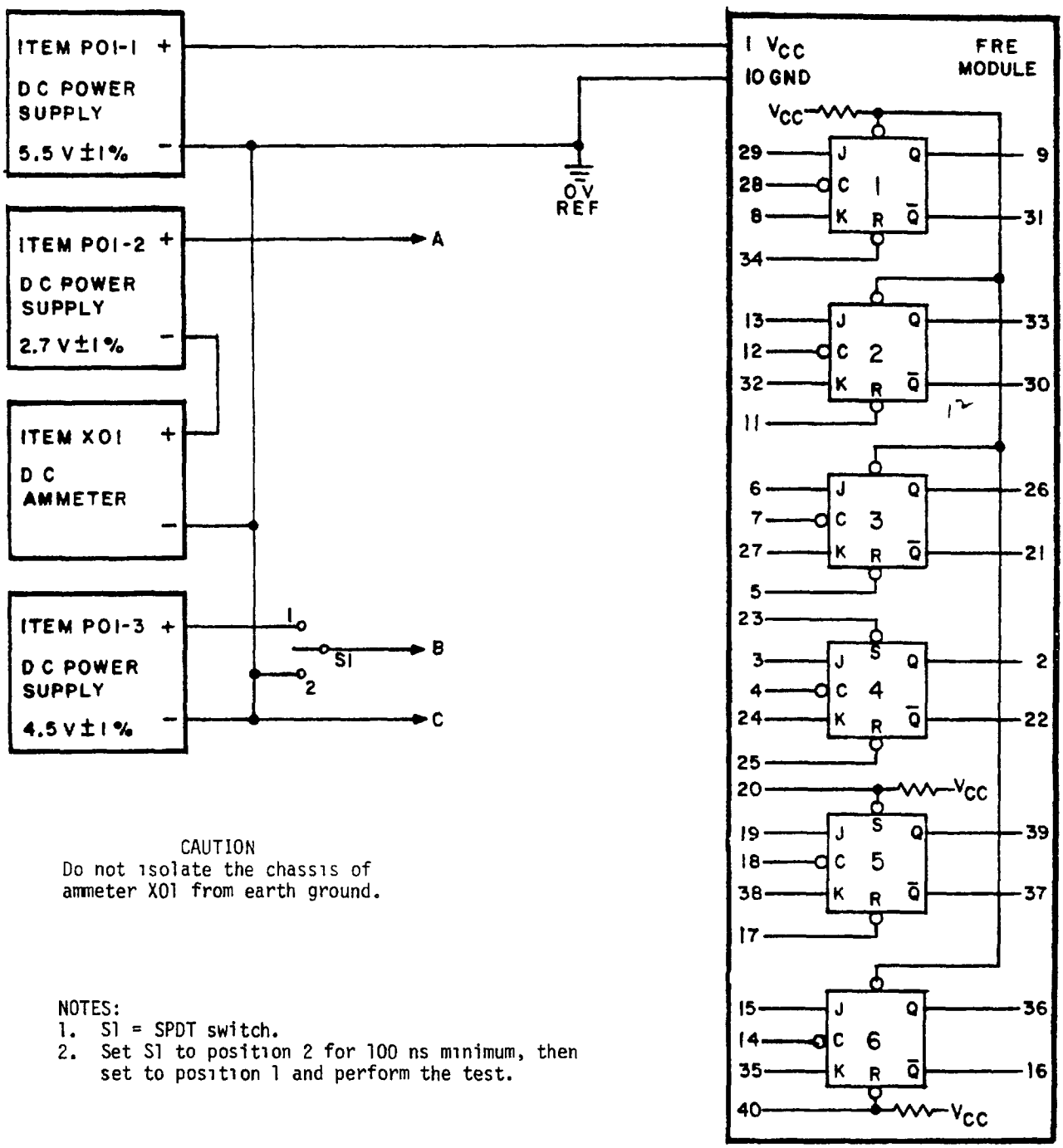
Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J1	1J	Input current, high level	29	Open	28,8,34	0	50	µA	12
J2	1K		8	34	29,28		100		
J3	2J		13	Open	12,32,11		50		
J4	2K		32	11	13,12		100		
J5	3J		6	Open	7,21,5		50		
J6	3K		27	5	6,7		100		
J7	4J		3	Open	4,24,25		50		
J8	4K		24	25	3,4		100		
J9	5J		19	Open	18,38,17		50		
J10	5K		38	17	19,18		100		
J11	6J		15	Open	14,35,40		50		
J12	6K		35	40	15,14		100		
J13	1C		28	Open	29,8,34				
J14	2C		12		13,32,11				
J15	3C		7		6,27,5				
J16	4C		4		3,24,25				
J17	5C		18		19,38,17				
J18	6C		14		15,35,4				
J19	1R		34		29,28,8				
J20	2R		11		13,12,32				

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TABLE X. Input current (high-level) test - continued.

Test condition	Parameter		Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
J21	3R	Input current, high level	5	Open	6,7,27	0	100	µA	12
J22	4R		25		3,4,24				
J23	5R		17		19,18,38				
J24	4S		23		3,4,24,25				
J25	5S		20		19,18,38,17	-166	-295		
J26	6R		40		15,14,35	-467	-626		

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CAUTION
Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPDT switch.
2. Set S1 to position 2 for 100 ns minimum, then set to position 1 and perform the test.

FIGURE 12. Input current (high-level) test circuit for table X.

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TABLE XI. Input current (low-level) test.

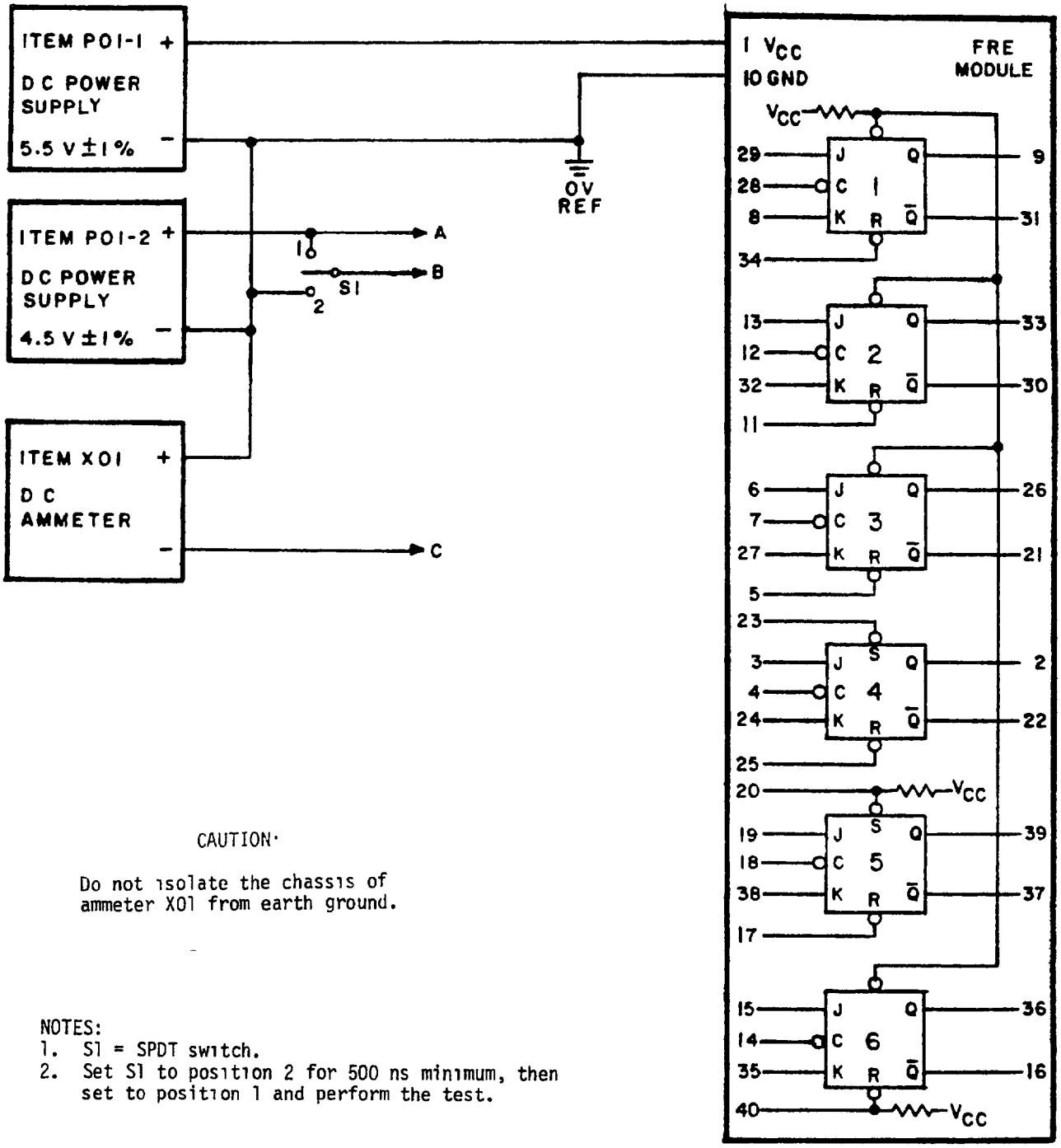
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K1	1J Input current, low level	28,34	Open	29	0.00	-1.77	mA	13
K2	1K	28,34		8				
K3	2J	12,11		13				
K4	2K	12,11		32				
K5	3J	7,5		6				
K6	3K	7,5		27				
K7	4J	4,25		3				
K8	4K	4,25		24				
K9	5J	18,17		19				
K10	5K	18,17		38				
K11	6J	14		15				
K12	6K	14		35		-7.73		
K13	1C	29,8	34	28				
K14	1R	29,28	Open	34				
K15	2C	13,32	11	12				
K16	2R	13,12	Open	11				
K17	3C	6,27	5	7				
K18	3R	6,7	Open	5				
K19	4C	3,24	25	4				
K20	4R	3,4	Open	25	0.00			

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TABLE XI. Input current (low-level) test - continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
K21	4S Input current, low level	4,24	Open	23	0.00	-7.73	mA	13
K22	5C	19,38	17	18	↓	-7.73	↓	↓
-								
K23	5R	19,18	Open	17		-7.73		
K24	5S	18,38	Open	20		-8.23		
K25	6C	15,35	40	14		-7.73		
K26	6R ↓	15,14	Open	40		-8.93		

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. S1 = SPDT switch.
2. Set S1 to position 2 for 500 ns minimum, then set to position 1 and perform the test.

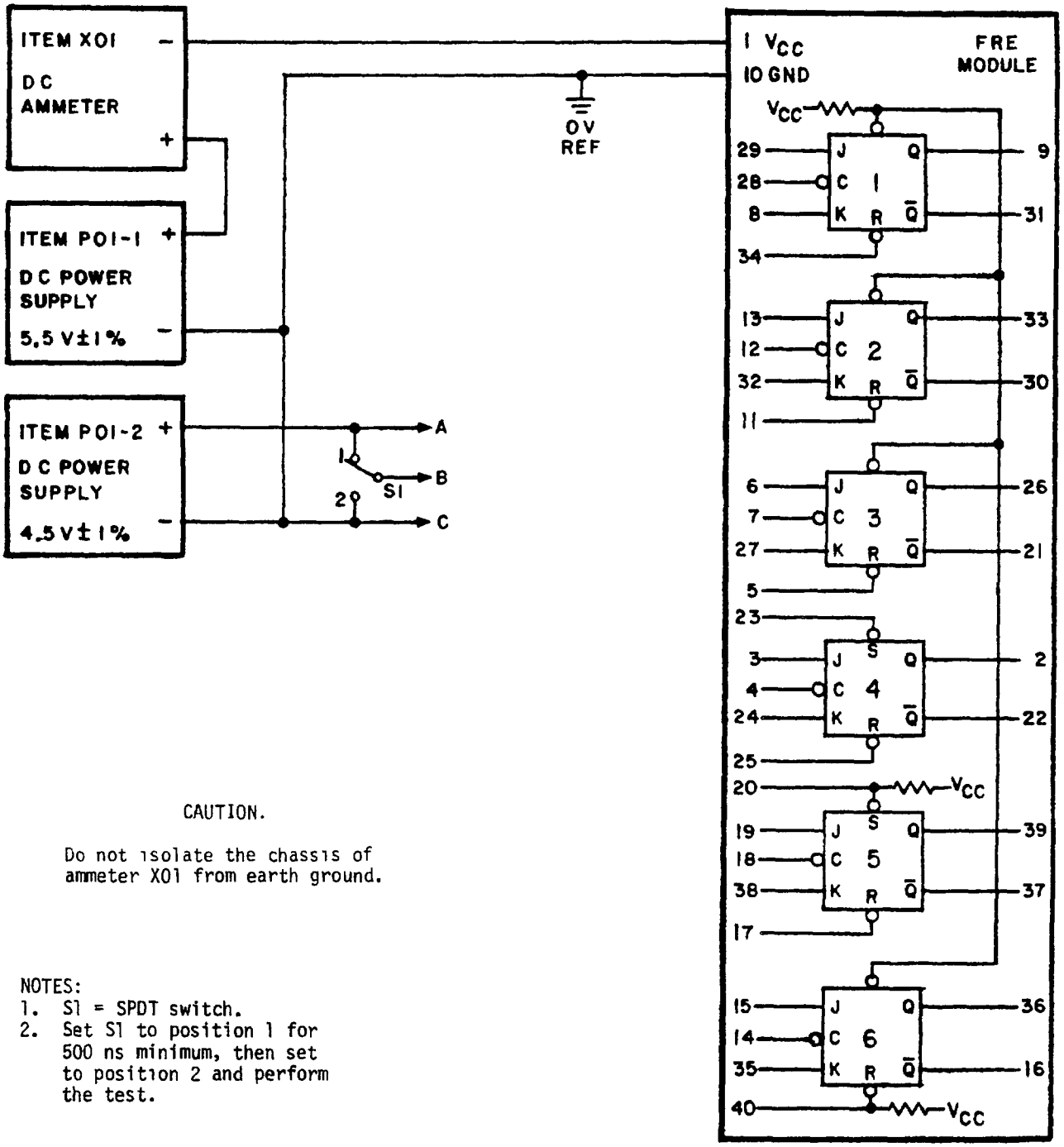
FIGURE 13. Input current (low-level) test circuit for table XI.

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TABLE XII. Power supply current test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	Power supply current	23	Open	29,28,8,34, 13,12,32,11, 6,7,27,5,3, 4,24,25,19, 18,38,17,15, 14,35,40	5	150	mA	14
L2	Power supply current	29,34,13,11, 6,5,3,25,19, 17,15	28,12,7, 14	8,32,27,24, 38,35,4,23, 18,20	5	150	mA	14

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CAUTION.

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. S1 = SPDT switch.
2. Set S1 to position 1 for 500 ns minimum, then set to position 2 and perform the test.

FIGURE 14. Power supply current test circuit for table XII.

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin pin	C to pin	Min	Max		
M1	1Q Propagation delay time (t_{PLH}), C_L = 50 pF	28	8,29,34	9	2	11	ns	15,16
M2	$1\overline{Q}$	28	8,29,34	31				
M3	2Q	12	13,32,11	33				
M4	$2\overline{Q}$	12	13,32,11	30				
M5	3Q	7	6,27,5	26				
M6	$3\overline{Q}$	7	6,27,5	21				
M7	4Q	4	3,24,25,23	2				
M8	$4\overline{Q}$	4	3,24,25,23	22				
M9	5Q	18	19,38,17	39				
M10	$5\overline{Q}$	18	19,38,17	37				
M11	6Q	14	15,35,40	36				
M12	$6\overline{Q}$	14	15,35,40	16				
M13	1Q Propagation delay time (t_{PLH}), C_L = 150 pF	28	8,29,34	9		21		
M14	$1\overline{Q}$	28	8,29,34	31				
M15	2Q	12	13,32,11	33				
M16	$2\overline{Q}$	12	13,32,11	30				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - continued.

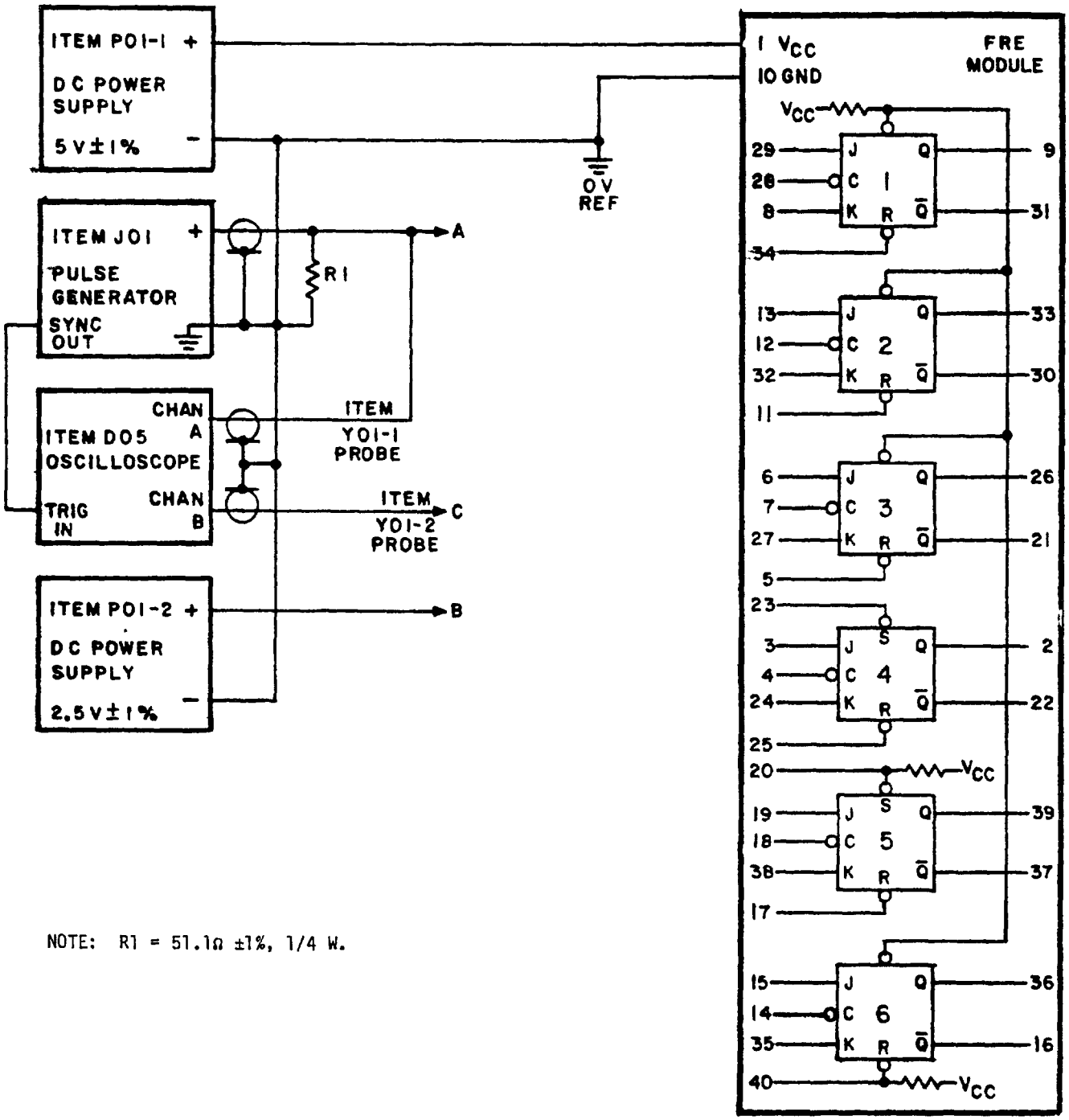
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin pin	C to pin	Min	Max		
M17	3Q Propagation delay time (t_{PLH}), C_L = 150 pF	7	6,27,5	26	2	21	ns	15,16
M18	$\overline{3Q}$	7	6,27,5	21				
M19	4Q	4	3,24,25,23	2				
M20	$\overline{4Q}$	4	3,24,25,23	22				
M21	5Q	18	19,38,17	39				
M22	$\overline{5Q}$	18	19,38,17	37				
M23	6Q	14	15,35,40	36				
M24	$\overline{6Q}$	14	15,35,40	16				
N1	1Q Propagation delay time (t_{PHL}), C_L = 50 pF	28	8,29,34	9		11		
N2	$\overline{1Q}$	28	8,29,34	31				
N3	2Q	12	13,32,11	33				
N4	$\overline{2Q}$	12	13,32,11	30				
N5	3Q	7	6,27,5	26				
N6	$\overline{3Q}$	7	6,27,5	21				
N7	4Q	4	3,24,25,23	2				
N8	$\overline{4Q}$	4	3,24,25,23	22				

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TABLE XIII. Propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin pin	C to pin	Min	Max		
N9	5Q Propagation delay time (t_{PHL}), C_L = 50 pF	18	19,38,17	39	2	11	ns	15,16
N10	$5\overline{Q}$	18	19,38 17	37				
N11	6Q	14	15,35,40	36				
N12	$6\overline{Q}$	14	15,35,40	16				
N13	1Q Propagation delay time (t_{PHL}), C_L = 150 pF	28	8,29,34	9		21		
N14	$1\overline{Q}$	28	8,29,34	31				
N15	2Q	12	13,32,11	33				
N16	$2\overline{Q}$	12	13,32,11	30				
N17	3Q	7	6,27,5	26				
N18	$3\overline{Q}$	7	6,27,5	21				
N19	4Q	4	3,24,25,23	2				
N20	$4\overline{Q}$	4	3,24,25,23	22				
N21	5Q	18	19,38,17	39				
N22	$5\overline{Q}$	18	19,38,17	37				
N23	6Q	14	15,35,40	36				
N24	$6\overline{Q}$	14	15,35,40	16				

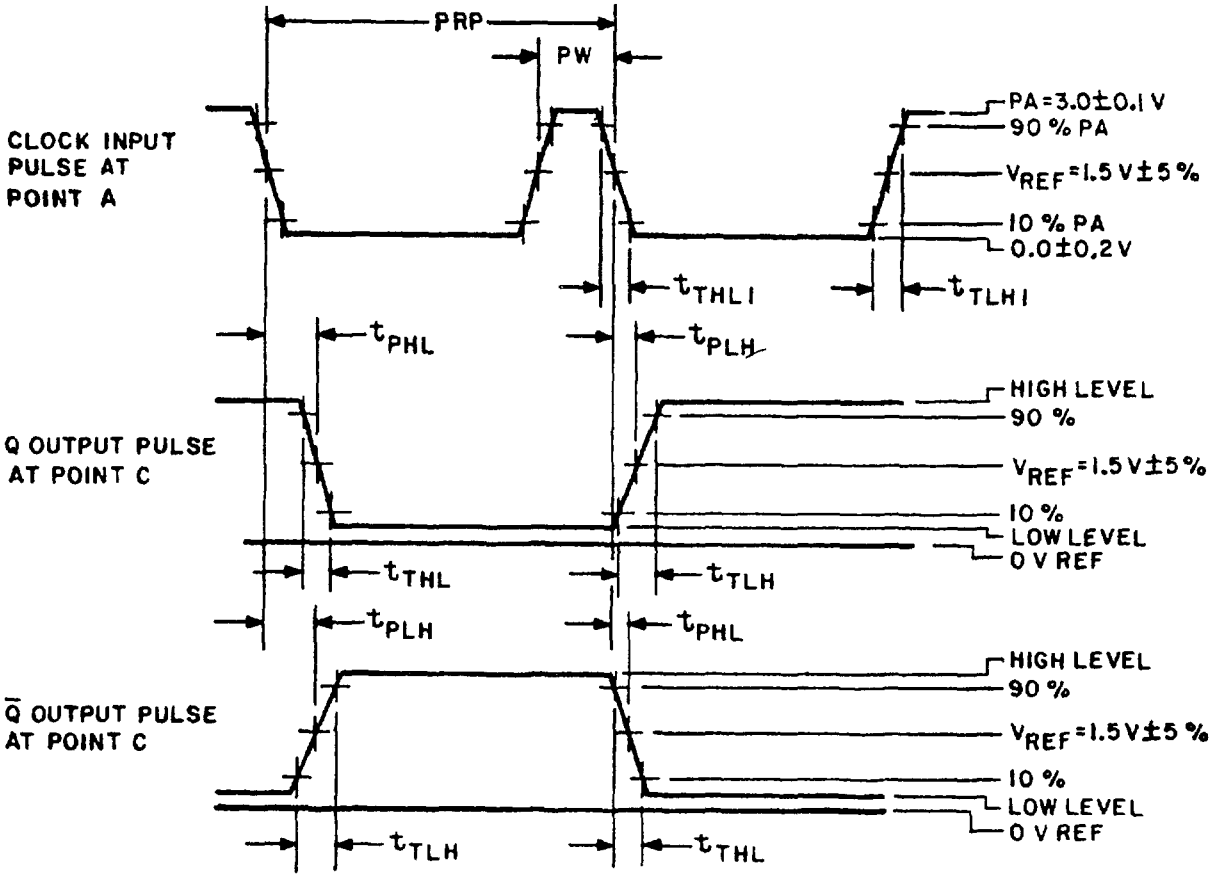
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 15. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIII.

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NOTES:
1. Input pulse parameters:
   PW = 500 ±50 ns.
   PRP = 1000 ±100 ns.
   $t_{TLH1} = t_{THL1}$ = 4 ±1 ns.
2. Verify output low level is $\leq$ 0.5 V and output high level is $\geq$ 2.5 V.

FIGURE 16. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIII.

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TABLE XIV. Setup time test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	1Q Setup time	29	28	9	8	900	1,100	ns	17,18
P2	2Q	13	12	33	32				
P3	3Q	6	7	26	27				
P4	4Q	3	4	2	24, 23				
P5	5Q	19	18	39	38				
P6	6Q	15	14	36	35				
P7	$1\overline{Q}$	8	28	31	29				
P8	$2\overline{Q}$	32	12	30	13				
P9	$3\overline{Q}$	27	7	21	6				
P10	$4\overline{Q}$	24	4	22	3, 23				
P11	$5\overline{Q}$	38	18	37	19				
P12	$6\overline{Q}$	35	14	16	15				
P13	$1\overline{Q}$	29	28	31	8				
P14	$2\overline{Q}$	13	12	30	32				
P15	$3\overline{Q}$	6	7	21	27				
P16	$4\overline{Q}$	3	4	2	24, 23				
P17	$5\overline{Q}$	19	18	37	38				
P18	$6\overline{Q}$	15	14	16	35				

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TABLE XIV. Setup time test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P19	1Q Setup time	8	28	9	29	900	1,100	ns	17,18
P20	2Q	32	12	33	13	↓	↓	↓	↓
P21	3Q	27	7	26	6				
P22	4Q	24	4	2	3, 23				
P23	5Q	38	18	39	19				
P24	6Q	35	14	36	15				

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TABLE XV. Hold time test.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q1	1Q	Hold time	29	28	9	8	900	1,100	ns	17,19
Q2	2Q		13	12	33	32				
Q3	3Q		6	7	26	27				
Q4	4Q		3	4	2	24, 23				
Q5	5Q		19	18	39	38				
Q6	6Q		15	14	36	35				
Q7	$1\overline{Q}$		8	28	31	29				
Q8	$2\overline{Q}$		32	12	30	13				
Q9	$3\overline{Q}$		27	7	21	6				
Q10	$4\overline{Q}$		24	4	22	3, 23				
Q11	$5\overline{Q}$		38	18	37	19				
Q12	$6\overline{Q}$		35	14	16	15				
Q13	$1\overline{Q}$		29	28	31	8				
Q14	$2\overline{Q}$		13	12	30	32				
Q15	$3\overline{Q}$		6	7	21	27				
Q16	$4\overline{Q}$		3	4	2	24, 23				
Q17	$5\overline{Q}$		19	18	37	38				
Q18	$6\overline{Q}$		15	14	16	35				

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TABLE XV. Hold time test - continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
Q19	1Q	Hold time	8	28	9	29	900	1,100	ns	17,19
Q20	2Q	↓	32	12	33	13	↓	↓	↓	↓
Q21	3Q		27	7	26	6				
Q22	4Q		24	4	2	3, 23				
Q23	5Q		38	18	39	19				
Q24	6Q	↓	35	14	36	15				

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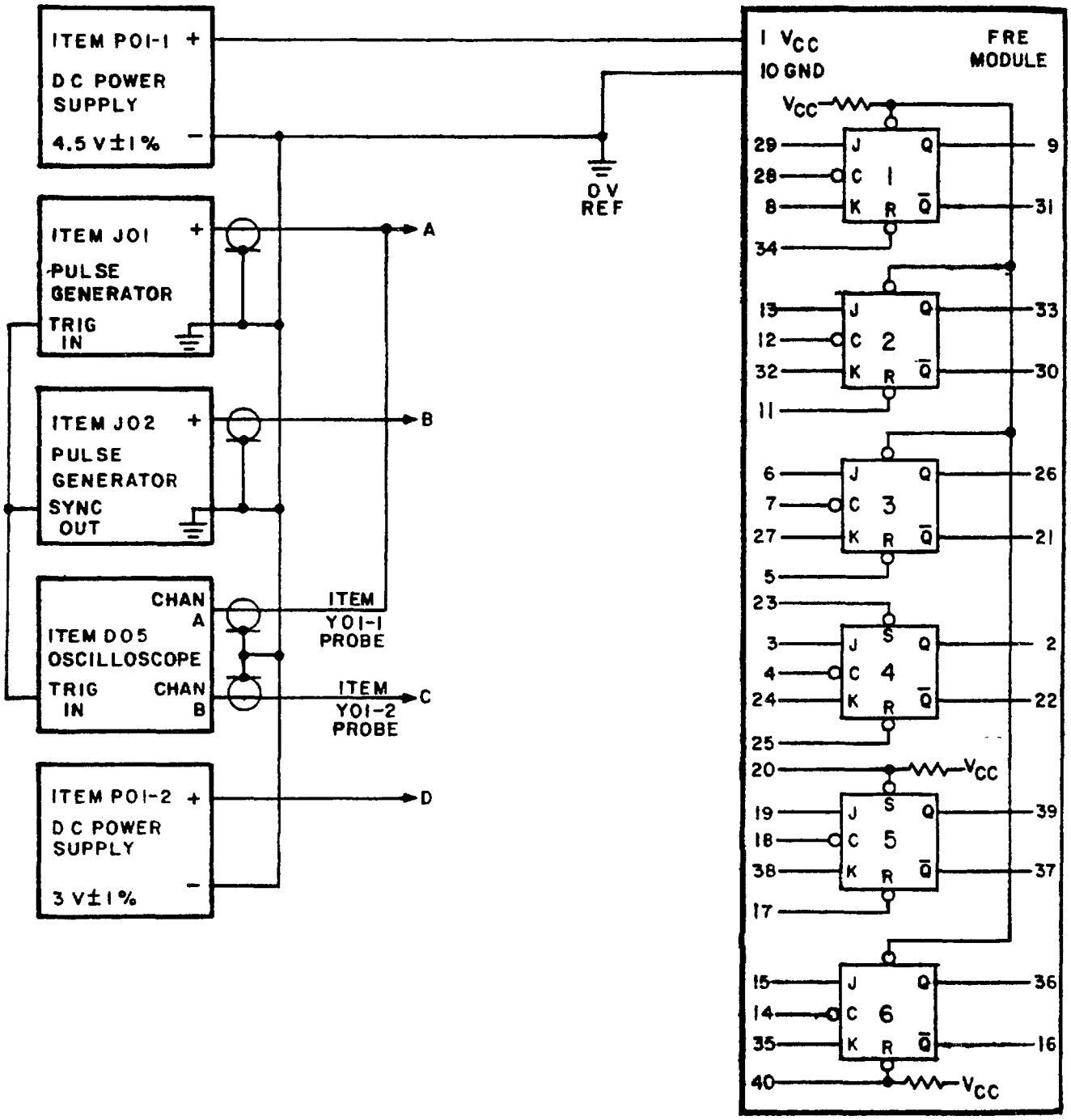
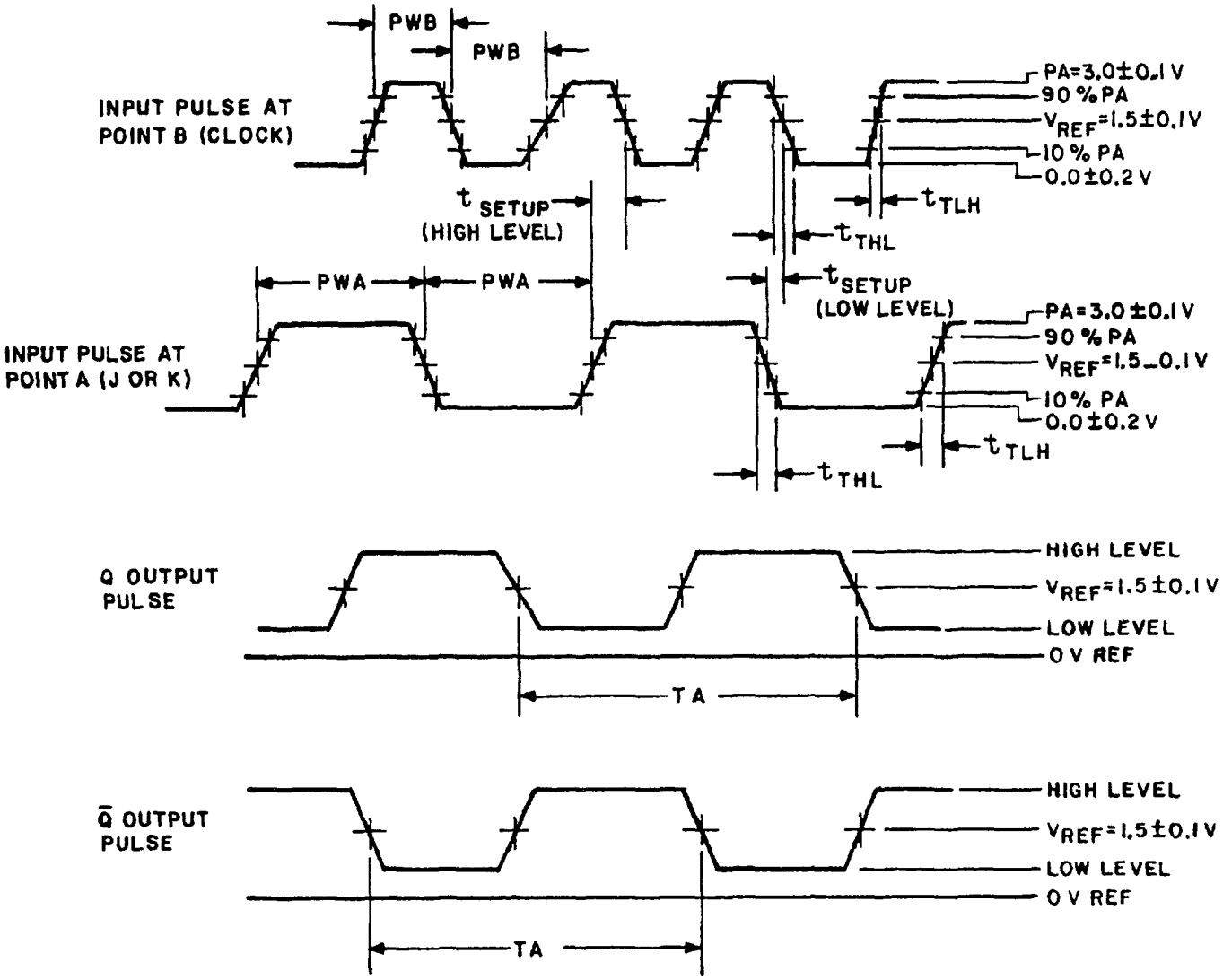


FIGURE 17. Setup and hold time test circuit for tables XIV and XV.

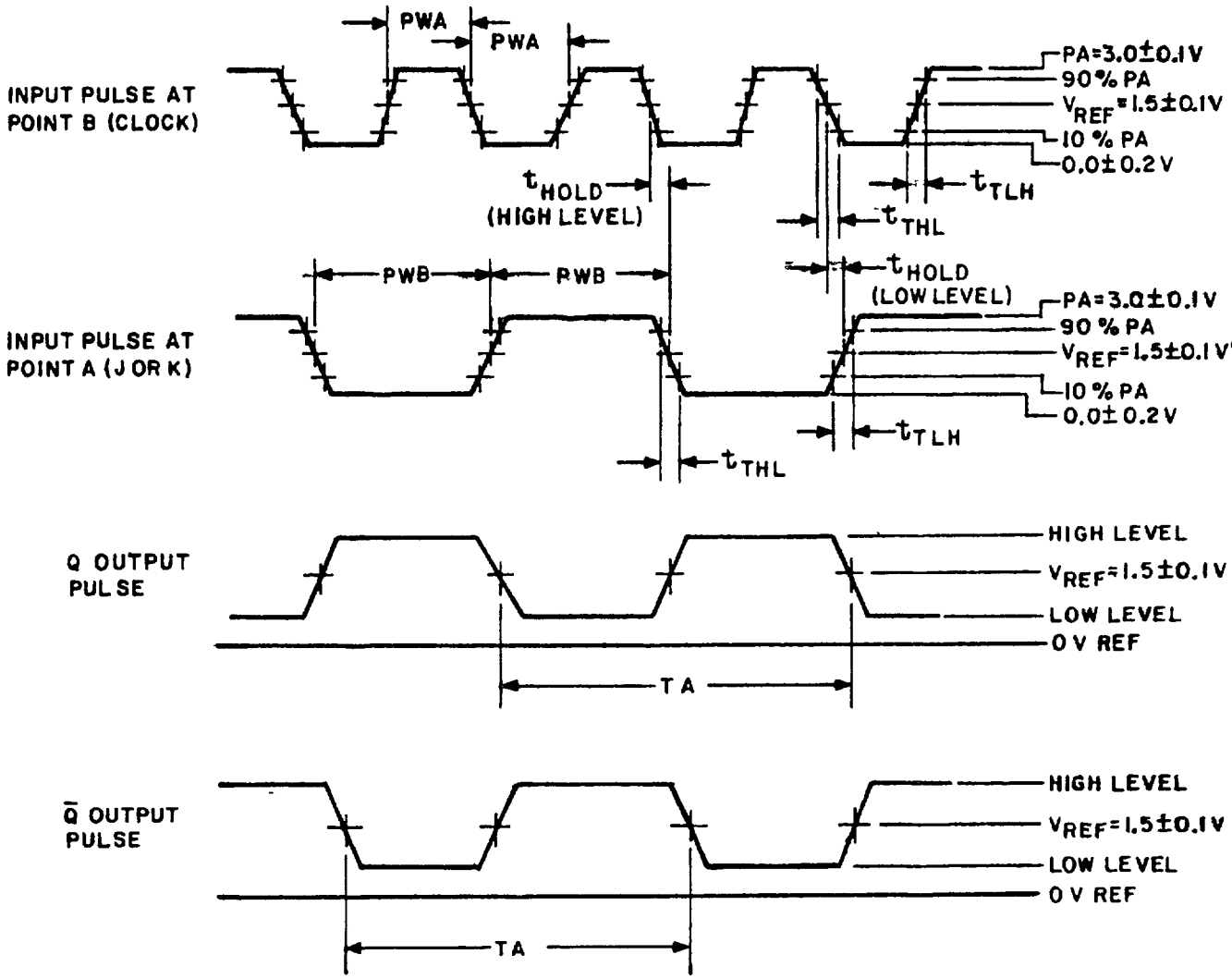
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NOTES:
1. Input pulse parameters:
   PWA = 500 ns ±10%.
   PWB = 250 ns ±10%.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   t_{SETUP} (HIGH LEVEL) = t_{SETUP} (LOW LEVEL) = 7 +0.2, -0 ns.
2. Measure TA by using item D05.
3. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.

FIGURE 18. Setup time test circuit waveforms for table XIV.

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NOTES:
1. Input pulse parameters:
   PWA = 250 ns ±10%.
   PWB = 500 ns ±10%.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   t_{HOLD} (HIGH LEVEL) = t_{HOLD} (LOW LEVEL) = 0 +5, -0 ns.
2. Measure TA by using item D05.
3. Before making the measurements, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.

FIGURE 19. Hold time test circuit waveforms for table XV.

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	S4 to Q4 Propagation delay time (t_{PLH}), C_L = 50 pF	25	4	23	2	2	11	ns	20,21
R2	S5 to Q5 Propagation delay time (t_{PLH}), C_L = 50 pF	17	18	20	39		11		
R3	S4 to Q4 Propagation delay time (t_{PLH}), C_L = 150 pF	25	4	23	2		21		
R4	S5 to Q5 Propagation delay time (t_{PLH}), C_L = 150 pF	17	18	20	39		21		
R5	R1 to $\overline{Q1}$ Propagation delay time (t_{PLH}), C_L = 50 pF	Open	28	34	31		11		
R6	R2 to $\overline{Q2}$	Open	12	11	30				
R7	R3 to $\overline{Q3}$	Open	7	5	21				
R8	R4 to $\overline{Q4}$	23	4	25	22				
R9	R5 to $\overline{Q5}$	Open	18	17	37				
R10	R6 to $\overline{Q6}$	Open	14	40	16	▼	▼	▼	▼

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - continued.

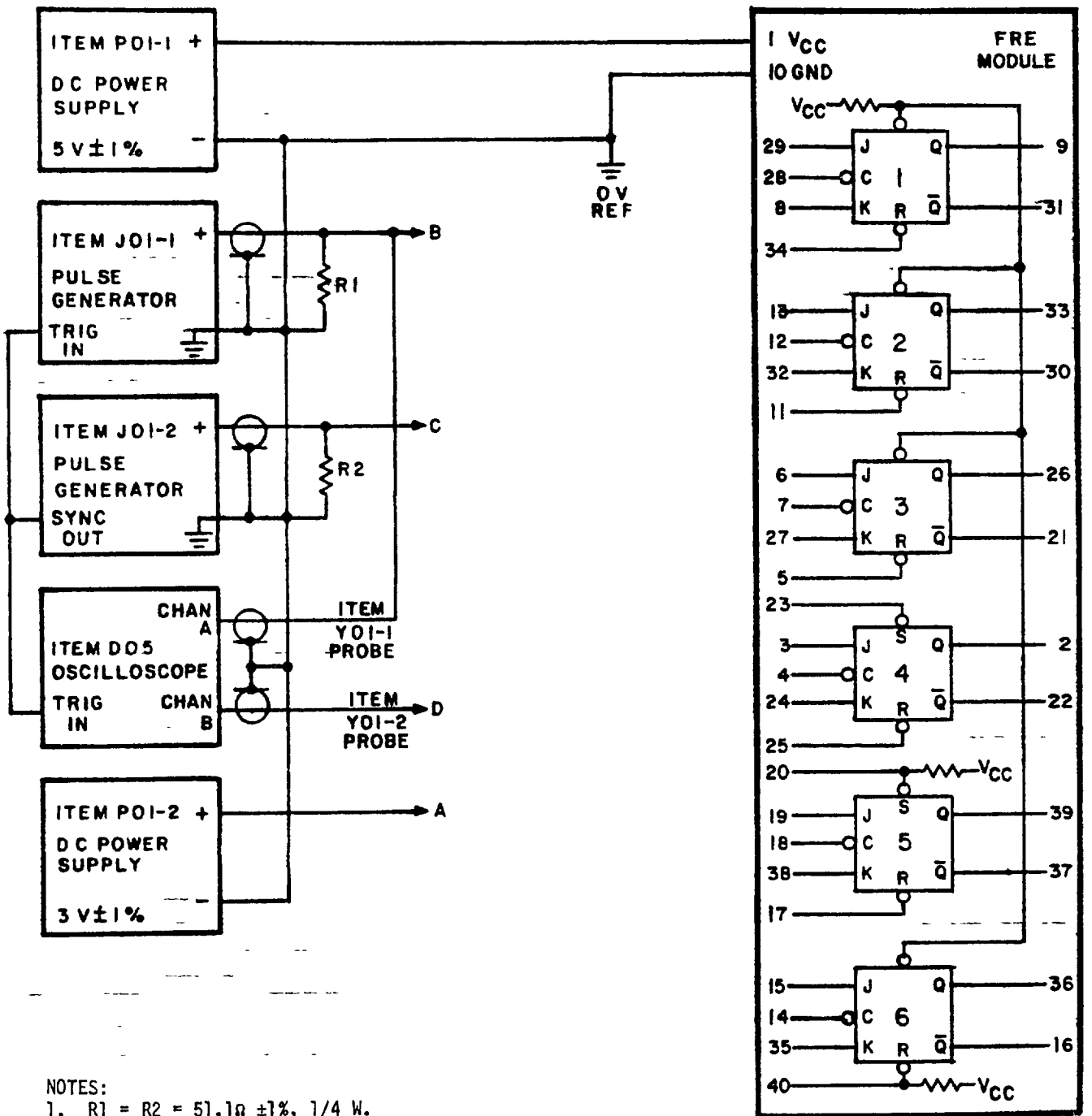
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
R11	R1 to $\overline{Q1}$	Propagation delay time (t_{PLH}), C_L = 50 pF	Open	28	34	31	2	21	ns	20,21
R12	R2 to $\overline{Q2}$	↓	Open	12	11	30	↓	↓	↓	↓
R13	R3 to $\overline{Q3}$	↓	Open	7	5	21	↓	↓	↓	↓
R14	R4 to $\overline{Q4}$	↓	23	4	25	22	↓	↓	↓	↓
R15	R5 to $\overline{Q5}$	↓	Open	18	17	37	↓	↓	↓	↓
R16	R6 to $\overline{Q6}$	↓	Open	14	40	16	↓	↓	↓	↓
S1	S4 to $\overline{Q4}$	Propagation delay time (t_{PHL}), C_L = 50 pF	25	4	23	22	↓	11	↓	↓
S2	S5 to $\overline{Q5}$	Propagation delay time (t_{PHL}), C_L = 50 pF	17	18	20	37	↓	11	↓	↓
S3	S4 to $\overline{Q4}$	Propagation delay time (t_{PHL}), C_L = 150 pF	25	4	23	22	↓	21	↓	↓
S4	S5 to $\overline{Q5}$	Propagation delay time (t_{PHL}), C_L = 150 pF	17	18	20	37	↓	21	↓	↓

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TABLE XVI. Propagation delay time (t_{PLH} and t_{PHL}) test - continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S5	R1 to Q1 Propagation delay time (t_{PHL}), C_L = 50 pF	Open	28	34	9	2	11	ns	20,21
S6	R2 to Q2	Open	12	11	33				
S7	R3 to Q3	Open	7	5	26				
S8	R4 to Q4	23	4	25	2				
S9	R5 to Q5	Open	18	17	39				
S10	R6 to Q6		14	40	36				
S11	R1 to Q1 Propagation delay time (t_{PHL}), C_L = 150 pF		28	34	9		21		
S12	R2 to Q2		12	11	33				
S13	R3 to Q3		7	5	26				
S14	R4 to Q4	23	4	25	2				
S15	R5 to Q5	Open	18	17	39				
S16	R6 to Q6	Open	14	40	36				

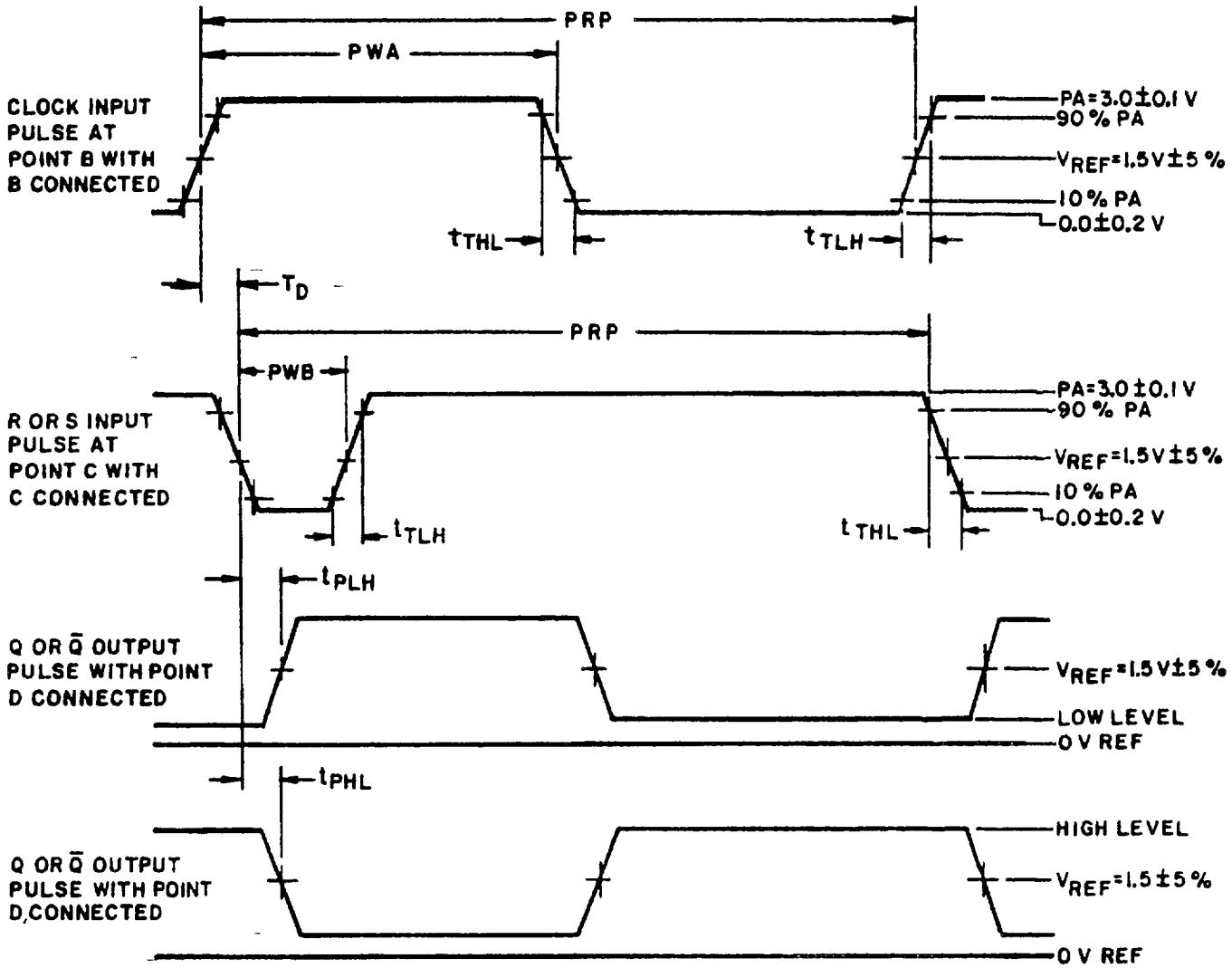
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load Q and $\overline{Q}$ outputs with load as specified on figure 11.

FIGURE 20. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PWB = 8 +2, -0 ns.
   PRP = 1000 ±100 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   $T_D = 50 \pm 5$ ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.5 V.
3. All input voltages are referenced to ground.

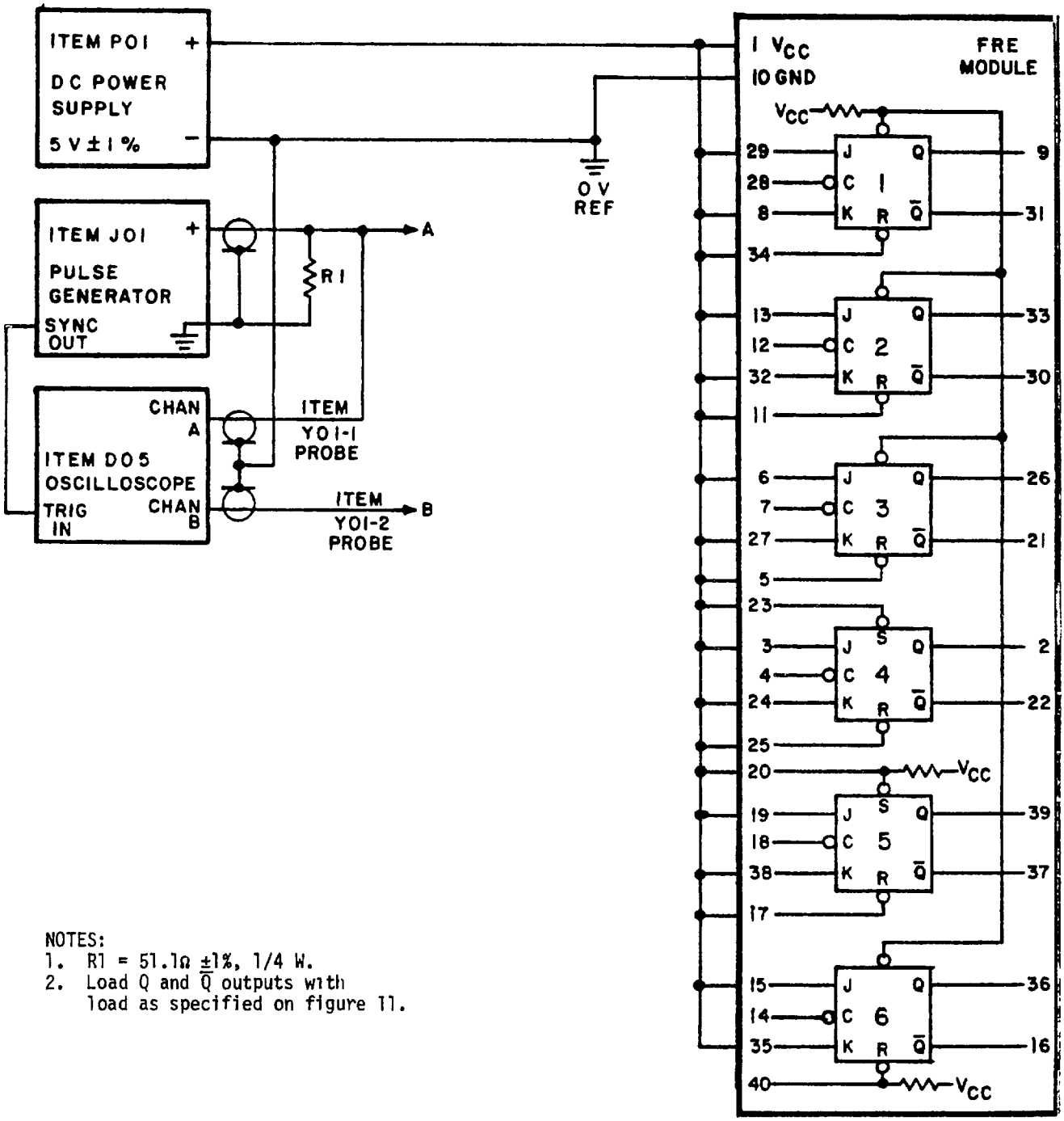
FIGURE 21. Propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Maximum clock frequency check test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
T1	Q1 Maximum clock frequency check	28	9	25	33	ns	22,23
T2	$\overline{Q1}$	28	31				
T3	Q2	12	33				
T4	$\overline{Q2}$	12	30				
T5	Q3	7	26				
T6	$\overline{Q3}$	7	21				
T7	Q4	4	2				
T8	$\overline{Q4}$	4	22				
T9	Q5	18	39				
T10	$\overline{Q5}$	18	37				
T11	Q6	14	36				
T12	$\overline{Q6}$	14	16				

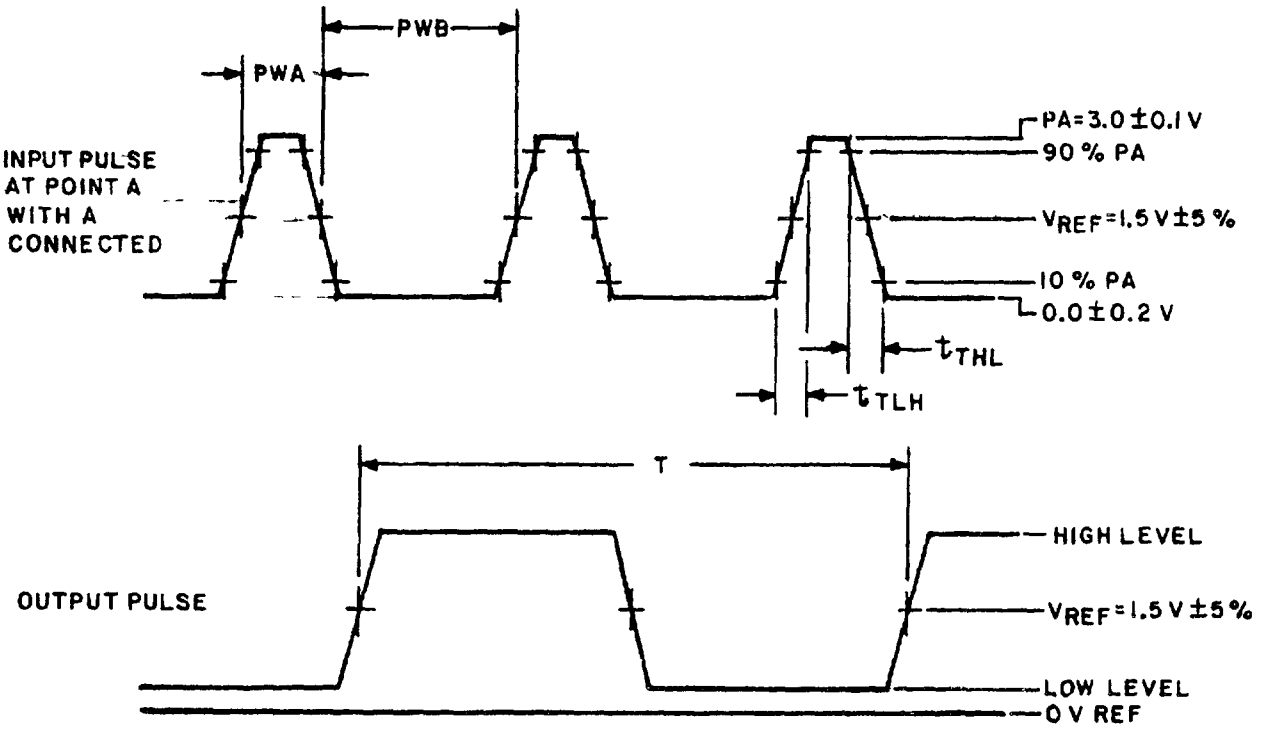
Source: https://assist.dla.mil -- Downloaded: 202640-20T17:40Z
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NOTES:
1. R1 = 51.1Ω ±1%, 1/4 W.
2. Load Q and $\overline{Q}$ outputs with load as specified on figure 11.

FIGURE 22. Maximum clock frequency check test circuit for table XVII.

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NOTES:
1. Input pulse parameters:
   PWA = 6 +2, -0 ns.
   PWB = 6.5 +2, -0 ns.
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 and the low level is $\leq$ 0.5 V.
3. All input voltages are referenced to ground.

FIGURE 23. Maximum clock frequency check test circuit waveforms for table XVII.

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TABLE XVIII. Output crosstalk (high-level and low-level) test.

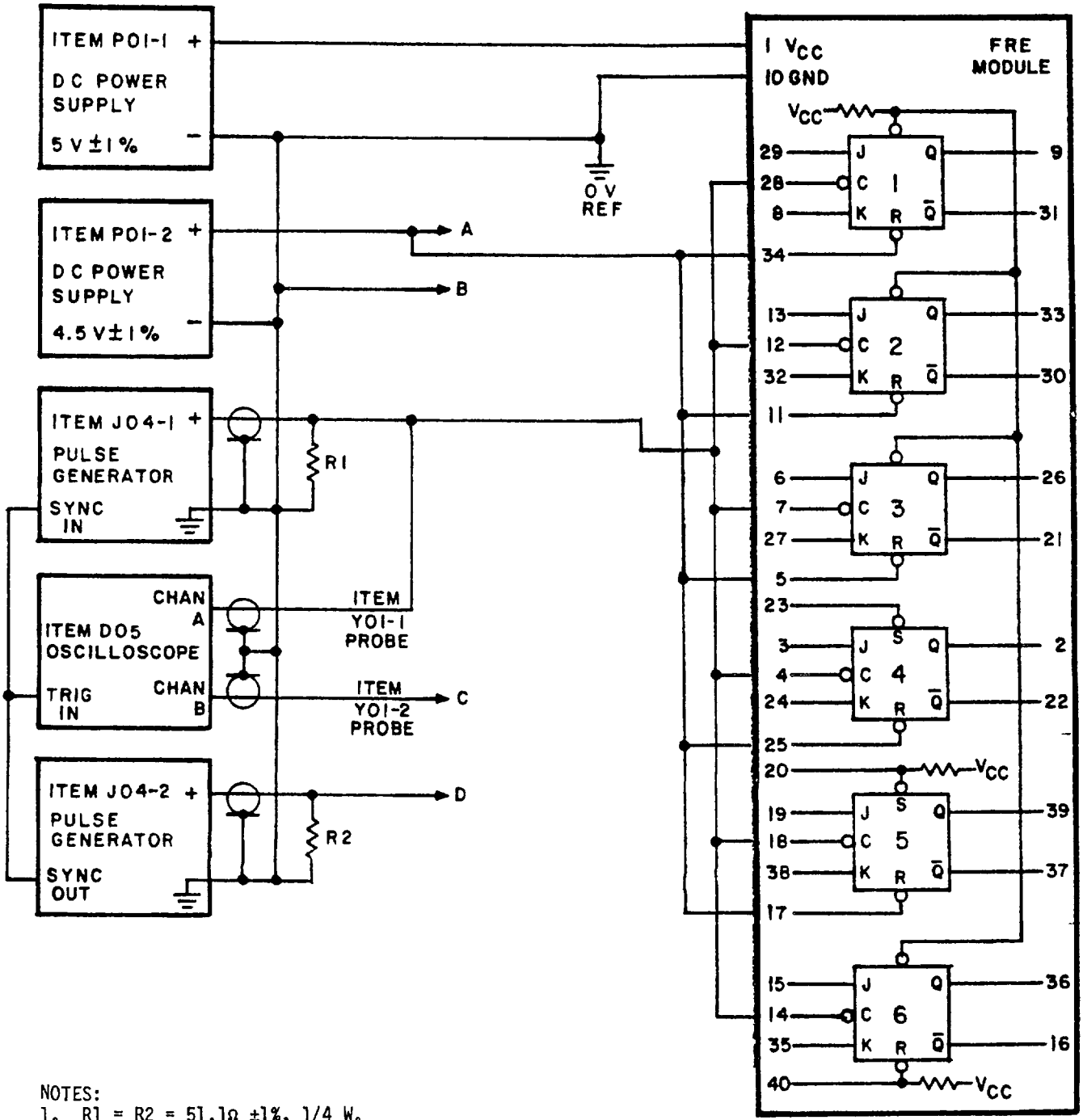
Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1Q	Output crosstalk, high level	29,32,27,24, 38 35	8	9	13,6,3,19,15	0	400	mV	24,25
U2	$1\overline{Q}$		8,32,27,24, 38,35	29	31	13,6,3,19,15				
U3	2Q		8,13,27,24, 38,35	32	33	29,6,3,19,15				
U4	$2\overline{Q}$		8,32,27,24, 38,35	13	30	29,6,3,19,15				
U5	3Q		8,32,6,24,38, 35	27	26	29,13,3,19,15				
U6	$3\overline{Q}$		8,32,27,24, 38,35	6	21	29,13,3,19,15				
U7	4Q		8,32,27,3,38, 35	24	2	29,13,6,19,15				
U8	$4\overline{Q}$		8,32,27,24, 38,35	3	22	29,13,6,19,15				
U9	5Q		8,32,27,24, 19,35	38	39	29,13,6,3, 15				
U10	$5\overline{Q}$		8,32,27,24, 38,35	19	37	29,13,6,3,15				
U11	6Q		8,32,27,24, 38,15	35	36	29,13,6,3,19				
U12	$6\overline{Q}$		8,32,27,24, 38,35	15	16	29,13,6,3,19				

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TABLE XVIII. Output crosstalk (high-level and low-level) test - continued.

Test condition	Parameter		Connect				Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
V1	1Q	Output crosstalk, low level	8,32,27,24, 38,35	29	9	13,6,3,19,15	0	500	mV	24,25
V2	$1\overline{Q}$	↓	29,32,27,24, 38,35	8	31	13,6,3,19,15	↓	↓	↓	↓
V3	2Q		8,32,27,24, 38,35	13	33	29,6,3,19,15				
V4	$2\overline{Q}$		8,13,27,24, 38,35	32	30	29,6,3,19,15				
V5	3Q		8,32,27,24, 38,35	6	26	29,13,3,19,15				
V6	$3\overline{Q}$		8,32,6,24,38, 35	27	21	29,13,3,19,15				
V7	4Q		8,32,27,24, 38,35	3	2	29,13,6,19,15				
V8	$4\overline{Q}$		8,32,27,3,38, 35	24	22	29,13,6,19,15				
V9	5Q		8,32,27,24, 38,35	19	39	29,13,6,3,15				
V10	$5\overline{Q}$		8,32,27,24, 19,35	38	37	29,13,6,3,15				
V11	6Q		8,32,27,24, 38,35	15	36	29,13,6,3,19				
V12	$6\overline{Q}$	↓	8,32,27,24, 38,15	35	16	29,13,6,3,19	↓	↓	↓	↓

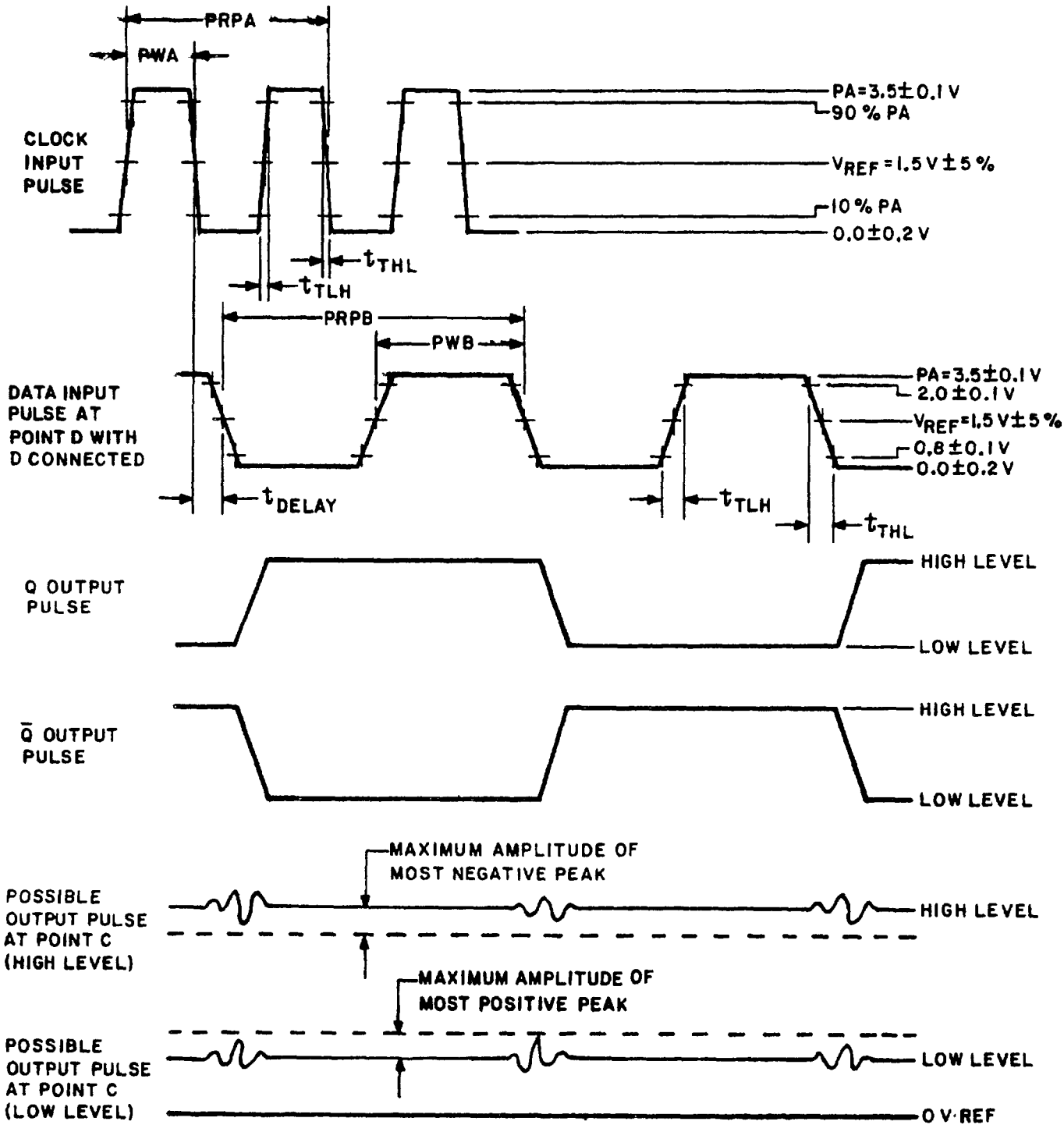
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NOTES:
1. R1 = R2 = 51.1Ω ±1%, 1/4 W.
2. Load Q and $\overline{Q}$ outputs with load as specified on figure 11.

FIGURE 24. Output crosstalk (high-level and low-level) test circuit for table XVIII.

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NOTE: Input pulse parameters:
PWA = 250 ±25 ns.
PWB = 500 ±50 ns.
PRPA = 500 ±50 ns.
PRPB = 1000 ±100 ns.
$t_{TLH} = t_{THL} = 10 \pm 1$ ns.
$t_{DELAY} = 125 \pm 12$ ns.

FIGURE 25. Output crosstalk (high-level and low-level) test circuit waveforms for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
W	Power supply transient amplitude	1	0	400	mV	26, 27

TABLE XX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	Power supply filter capacitance	1	Open	1.6	3.0	µF	28

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TABLE XXI. Input capacitance test.

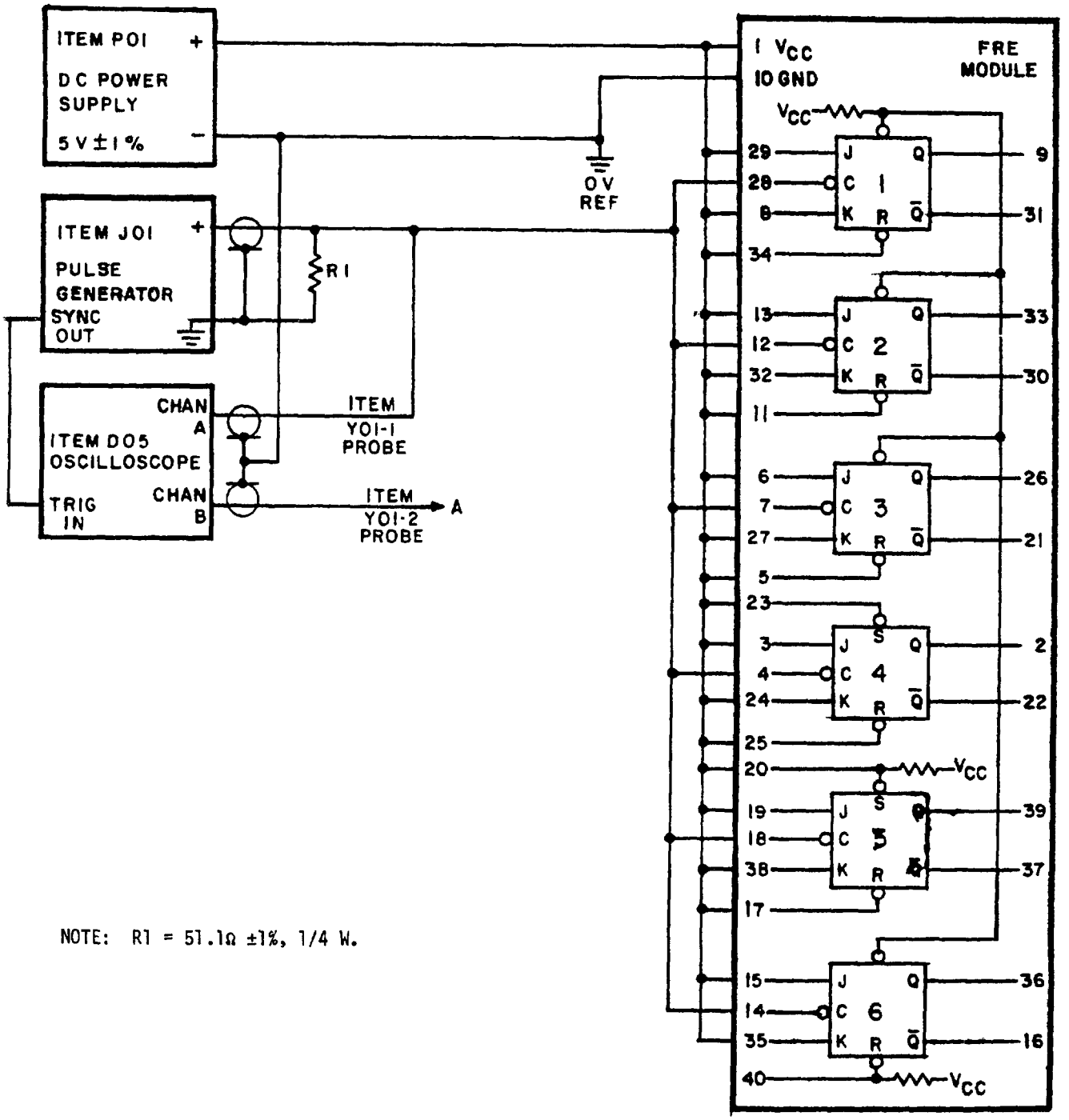
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	1J Input capacitance	Open	29	2	18	pF	28
Y2	1K		8				
Y3	2J		13				
Y4	2K		32				
Y5	3J		6				
Y6	3K		27				
Y7	4J		3				
Y8	4K		24				
Y9	5J		19				
Y10	5K		38				
Y11	6J		15				
Y12	6K		35				
Y13	1C		28		25		
Y14	1R		34				
Y15	2C		12				
Y16	2R		11				
Y17	3C		7				
Y18	3R		5				
Y19	4C		4				
Y20	4S		23				

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TABLE XXI. Input capacitance test continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
Y21	4R	Input capacitance	Open	25	2	25	pF	28
Y22	5C	↓	↓	18	↓	↓	↓	↓
Y23	5S			20				
Y24	5R			17				
Y25	6C			14				
Y26	6R			40				

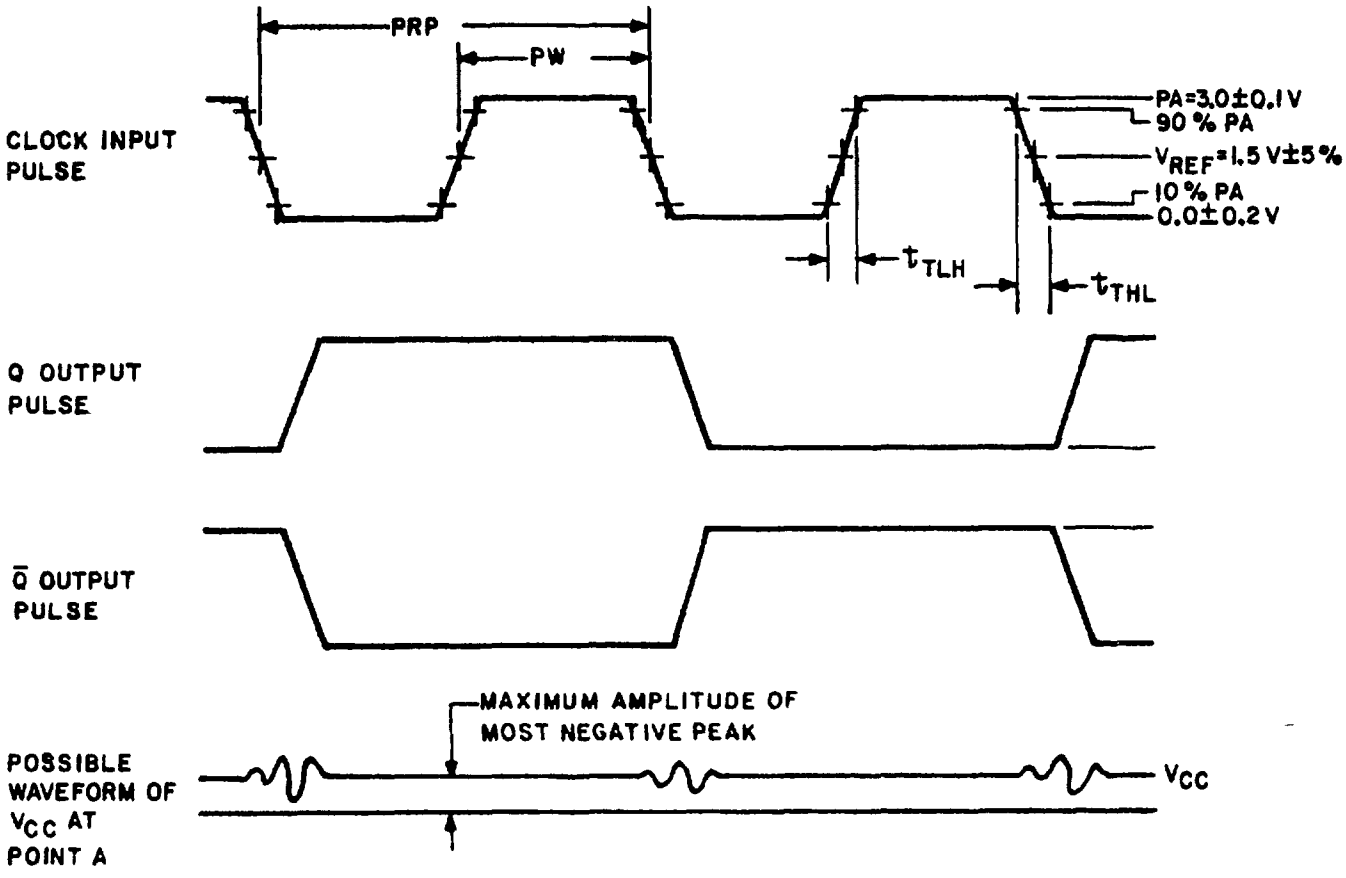
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NOTE: R1 = 51.1Ω ±1%, 1/4 W.

FIGURE 26. Power supply transient amplitude test circuit for table XIX.

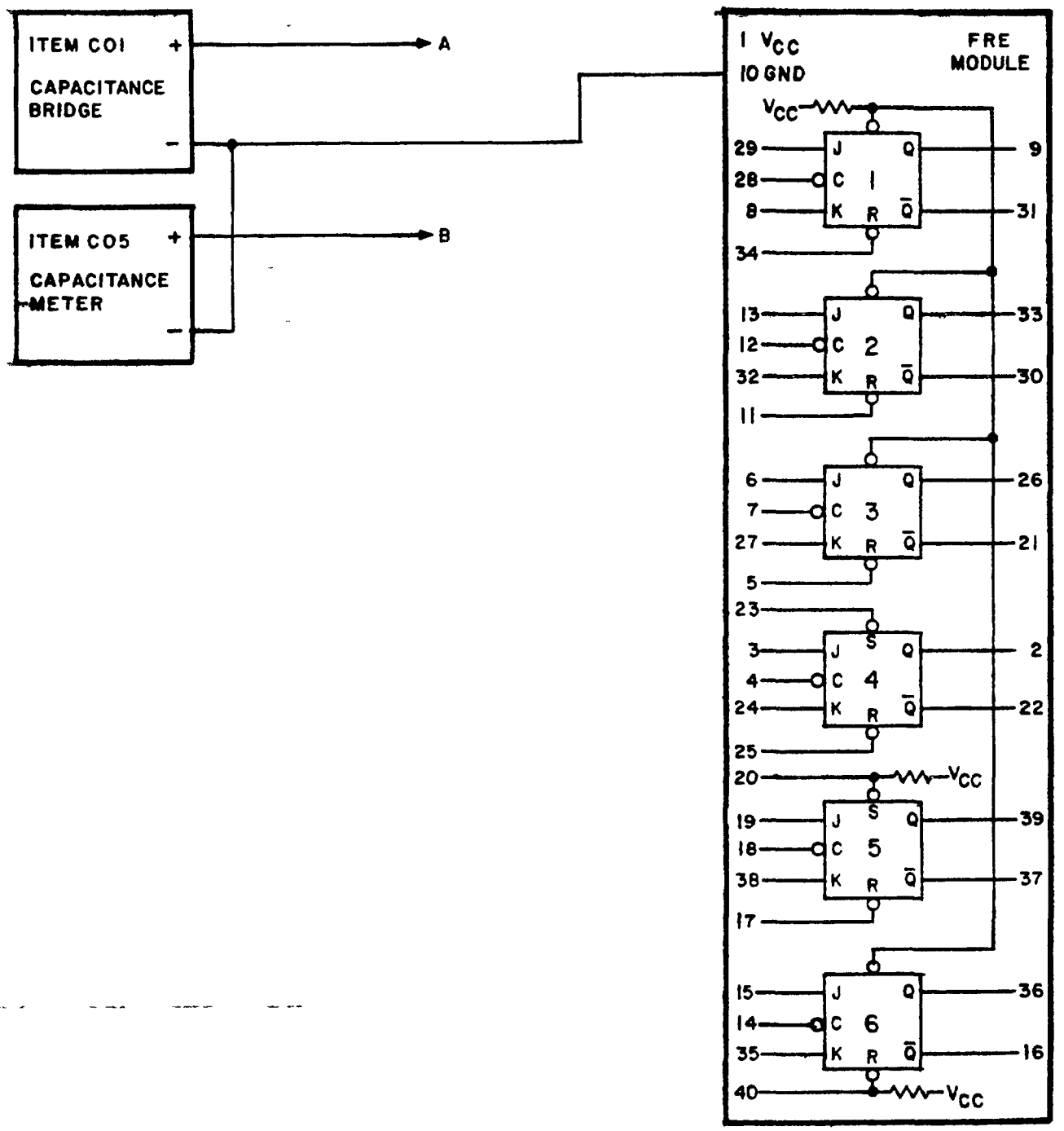
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NOTE: Input pulse parameters:
PW = 500 ±50 ns.
PRP = 1000 ±100 ns.
$t_{TLH} = t_{THL}$ = 10 ±2 ns.

FIGURE 27. Power supply transient amplitude test circuit waveforms for table XIX.

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NOTES:
1. Null out the effect of the test leads prior to connection to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 28. Power supply filter capacitance and input capacitance test circuit for tables XX and XXI.

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